National Comprehensive Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

# Immunotherapy Side Effects: Immune Checkpoint Inhibitors

Presented with support from NATIONAL COMPREHENSIVE CANCER NETWORK® FOUNDATION Guiding Treatment. Changing Lives.

Available online at NCCN.org/patientguidelines

# NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

**These NCCN Guidelines for Patients are based on the NCCN Guidelines® for Management of Immune Checkpoint Inhibitor-Related Toxicities, Version 1.2026 — October 23, 2025**

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# 1

# About immune checkpoint inhibitors and side effects


5 What is immune checkpoint inhibitor therapy?

5 How do immune checkpoint inhibitors work?

7 What do I need to know about immune checkpoint inhibitor side effects?

7 How are these side effects treated?

7 How to use this guide

8 How can I get the best care?

If you are having immunotherapy treatment for cancer with immune checkpoint inhibitors (ICIs), it's important to know about side effects you may have. Sometimes there are symptoms with these side effects and sometimes there aren't. Some side effects can only be found using tests. These include blood and imaging tests.

## What is immune checkpoint inhibitor therapy?

Immune checkpoint inhibitor (ICI) therapy is a type of immunotherapy. Immunotherapy is a treatment that uses the body's immune system to attack cancer cells. The immune system fights disease and infection. For a list of ICIs, see **Guide 1**.

In the past, ICI therapy was used to treat cancer at advanced stages. But it has become an option for treating earlier stages. It's used to treat many types of cancer.

The main types of ICIs are:

- PD-1/PD-L1 inhibitors
- CTLA-4 inhibitors
- LAG-3 inhibitors

## How do immune checkpoint inhibitors work?

They help the immune system find and attack cancer cells.

Here's how: There are proteins on the surface of cancer cells (such as PD-L1) that interact with proteins on immune cells (such as PD-1). When thse proteins interact, the immune cell doesn't see the cancer cell as a danger. ICIs prevent this interaction, so the immune cell can do its job—attack the cancer cell.

**Guide 1**
**Immune checkpoint inhibitors**

**PD-1/PD-L1 inhibitors**

Cemiplimab (Libtayo)
Nivolumab (Opdivo)
Pembrolizumab (Keytruda)
Atezolizumab (Tecentriq)
Avelumab (Bavencio)
Durvalumab (Imfinzi)
Dostarlimab (Jemperli)
Retifanlimab (Zynyz)
Tislelizumab (Tevimbra)

**CTLA-4 inhibitors**

Ipilimumab (Yervoy)
Tremelimumab (Imjudo)

**LAG-3 inhibitor**

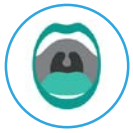
Relatlimab (Opdualag)

## Immune checkpoint inhibitor therapy side effects

**Side effects can affect any part of the body.**

### Head and neck

**Eyes**
Blurred or distorted vision, blind spots, eye pain or redness, itchy or bulging eyes



**Mouth**
Dry mouth, pain or burning, sores, difficulty eating

**Brain or spinal cord**
Headaches, sensitivity to light, confusion, changes in behavior

**Thyroid**
Weight loss or gain, trouble pooping, dry skin, sensitivity to cold

**Pituitary gland**
Fatigue (caused by low cortisol), loss of appetite, nausea, dizziness

### Chest

**Lungs**
Trouble breathing, dry cough, fever, chest pain

**Heart**
Chest pain, irregular heartbeat, shortness of breath, fatigue, weakness

### Abdomen (belly)

**Kidneys**
Difficulty filtering waste products from blood and pee

**Liver**
Too much bilirubin in bile, and certain enzymes in liver

**Pancreas**
Nausea, bloating, belching, belly or back pain, vomiting, abnormal heart rate

**Stomach and bowel**
Nausea, stomach pain or cramps, fever, watery or loose poop, blood or mucus in poop

### Whole body

**Skin**
Rash, itching, blisters, itchy bumps and patches, scaly skin

**Nerves**
Muscle weakness, weakness in face, difficulty breathing

**Joints**
Joint pain, swelling, stiffness

**Muscles**
Muscle aches, pain, stiffness

**Blood**
Low red blood cells, low blood platelets, low white blood cells

## What do I need to know about immune checkpoint inhibitor side effects?

Not all side effects cause symptoms. The following are some important things to know about ICI therapy side effects:

- They may start shortly after you begin treatment, within the first couple of months, or even after treatment is finished.
- You're more likely to have side effects if you're taking more than 1 kind of immunotherapy or an immunotherapy combined with other types of cancer medicines, like chemotherapy.
- Many of these side effects are mild and can be treated. But there are some rare and serious side effects that can lead to death, especially if untreated.
- Some side effects are chronic. This means they may need to be managed for the rest of your life.
- It's important to be aware of possible side effects and contact your health care team if you have symptoms.
- Side effects can affect 1 organ or body system or several organs or body systems.

## How are these side effects treated?

Side effects are often treated with steroids, but treatment depends on many things, including how severe the side effects are and how much they are affecting your life.

## How to use this guide

Here is an overview of how you can use this resource: In *Chapter 2: What to know before immune checkpoint inhibitor therapy*, you can learn about tests you may have before starting treatment. A physical exam with a personal and family medical history will help your care team make choices about your treatment. You can read about the kinds of medicines you may be given if you have side effects because of treatment.

Chapters 3 through 12 cover the side effects that may affect the body's organs or systems. Each chapter explains:

- The body system or organs that may be affected
- Tests or procedures you may have if you're having side effects
- How the side effects will be managed
- If and how side effects will affect your immunotherapy program

Chapter 13 discusses other resources you may find helpful.

## How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters.

Keep reading to find the best care for you.

### How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team.

# 2

# What to know before immune checkpoint inhibitor therapy


10 Preparing for immune checkpoint inhibitor therapy

12 About managing side effects

16 Fatigue: a common side effect

17 Communication is key

18 Key points

18 Questions to ask

This chapter explains what will happen and how you can prepare for immune checkpoint inhibitor (ICI) therapy.

You'll be monitored for side effects during immune checkpoint inhibitor (ICI) treatment. Tell your care team if you have any symptoms so they can help you manage them.

Fatigue, or extreme tiredness, is a common side effect of ICI therapy. But fatigue can also be due to other illnesses or conditions. In this chapter, you'll read about ICI side effects that can cause fatigue and how to find out more about what may be causing it.

## Preparing for immune checkpoint inhibitor therapy

Before you have ICI therapy, your health care team will want to find out as much as possible about your personal and family medical history. You'll have a physical exam before starting treatment.

### Medical history

A medical history is a record of all health issues and treatments you have had in your life. This includes:

- Illness or injuries
- Medicines you take or have taken, including supplements, prescriptions, and over-the-counter medicine

It's important to tell your care team if you have received an organ transplant or hematopoietic cell cell transplant (HCT).

### Family history

A family history is a record of the health of your family members who are blood relatives. Your health care team will ask about your history or any family member's history of:

- Hormone issues
- Disease, including:
  - Autoimmune disease, like type 1 diabetes and rheumatoid arthritis
  - Organ-specific disease, like heart or lung disease

### Physical exam

During a physical exam, you will:

- Have blood tests (for information about blood sugar and blood cell counts, and how well organs, such as the liver and kidneys, are working)
- Have your pulse and breathing rate checked
- Be asked about your bowel habits, how often you usually poop, and the typical shape and texture of your bowel movements (poops)
- Have your thyroid hormone and cortisol hormone tested

You may:

- Have imaging tests, which may include an MRI of the brain, chest x-ray, or CT scans
- Be screened for HIV; hepatitis A, B, and C; or all of these diseases
- Have your skin and mucous membranes checked if you have a history of immune-related skin issues

- Have heart enzyme tests and an electrocardiogram (ECG) to check electrical activity in the heart
- Have a neurologic exam to check your mental status, reflexes, and movement if you've had a nervous system (affecting the brain, spine, or nerves) issue in the past

You'll have blood tests before each treatment and throughout your treatment program to learn if the cancer is responding to treatment and if you're having any side effects.

## Other things to know before treatment

It's important to let your primary care team and all of your health care team know you're having—or have had in the past—ICI treatment.

### Fertility and breastfeeding

The effect of immunotherapy on fertility—people's ability to have children—is unknown. So, use of birth control is advised for people of childbearing age during immunotherapy and for at least 5 months after the last dose. The same recommendation is made for breastfeeding.

You may want to discuss family planning and fertility with your care team, if these are concerns you have. You may be referred to a

**Prescription medicines are ordered by a doctor.**

**Over-the-counter medicines can be bought without a doctor's prescription.**

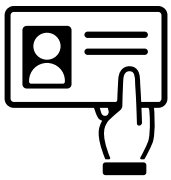
### Vaccines

Inactivated, or killed, vaccines are safe during immunotherapy. Activated (live) ones are not recommended. If you aren't sure whether it's safe for you to get a vaccine, talk to your health care team.

## How is ICI therapy given?

ICI therapy can be given as an infusion injected with a needle into a vein. This is called intravenous, or IV.

In some cases, ICI therapy may be given into the skin with a needle smaller than an IV needle. This is called subcutaneous.



**Immunotherapy Wallet Card**

Ask your care team for an immunotherapy wallet card. This card lists your immunotherapy regimen, potential side effects, and contact numbers for your cancer care team. Carry it with you at all times. If a card is not available, ask for a printable list of your treatment regimen.
**A printable card is available from the Oncology Nursing Society (ONS) website.**

### Infusion-related side effects

Some people have reactions during the infusion itself. Most are mild.

Some ICIs are more likely than others to cause reactions when they are given. Reactions can include:

- Low-grade fever
- Chills
- Headache
- Nausea

Sometimes medicine is given before infusion to help prevent severe reactions.

## About managing side effects

In some cases, the first step in managing a side effect is to pause or stop the immunotherapy treatment. This usually stops the side effect, too.

Your care team will work with specialized experts as needed to guide your care. Some severe reactions may require treatment at a focused care center.

If your ICI treatment has been paused because you've had a severe side effect, your care team will be careful when deciding whether to re-start treatment. They will assess how well the cancer responded to decide if they think it's a good idea to start again. They will discuss with you the risks and benefits of restarting ICI therapy.

### Steroids

Many side effects of ICIs are treated with steroids. Steroids reduce the activity of the immune system and reduce inflammation.

- Prednisone and methylprednisolone are the most commonly used steroids.
  - Prednisone is taken orally, as a pill that is swallowed.
  - Methylprednisolone is injected into the bloodstream by IV.
- Some steroids can be put on the skin to treat skin or mouth issues. These are called topicals.
- Certain side effects of the eyes can be treated with steroid eye drops.

Steroids work best when they are started early—when symptoms or other issues first

come up. If you notice new symptoms, or the symptoms you have are getting worse, talk to your care team right away.

### Side effects of steroids

Steroids can be helpful in managing treatment side effects, but they can also cause side effects. Steroids can:

- **Irritate the stomach lining.** If you take other medicines that raise your risk for stomach-lining issues, you may also be given another medicine to help prevent stomach issues.
- **Lead to osteoporosis.** Steroids used for a long time can lead to osteoporosis. Osteoporosis is when bones get weak. If you will be taking steroids for a long time, vitamin D and calcium will be added to your treatment. Your care team will also recommend physical therapy, seeing an endocrinologist (a hormone expert), and doing weight-bearing exercises.
- **Lead to infections.** If steroids are taken at high doses for a long time, an infection is more likely to occur. If this is the case, your care team may prescribe an antibiotic.

Keep in mind: Stopping steroids too quickly can cause side effects to return. That's why steroid doses are lowered slowly over time. This is called tapering.

## Other kinds of treatments that may be used

Not all side effects are managed with steroids. Other types of treatment include:

- Biologics
  - Intravenous immunoglobulin (IVIG)
  - Tumor necrosis factor (TNF) blockers
  - Other immune blockers
- Gabapentinoids
  - Gabapentin
  - Pregabalin
- Hormones *(Chapter 7: Side effects of the endocrine system)*
- Methotrexate
- Mycophenolate mofetil

Some of these treatments are discussed next.

## Biologics

Biologics are medicines made from living things like proteins and genes. There are many kinds of biologic medicines that can help prevent and treat a wide range of diseases. Most biologics are given by IV.

IVIG is a biologic used to fight infection. It can also be used to treat ICI side effects related to the:

- Skin (see Chapter 3)
- Nerves (see Chapter 10)
- Muscles and joints (see Chapter 11)
- Lungs and heart (see Chapter 12)
- Blood and bone marrow (see Chapter 12)

Some biologics reduce inflammation. Inflammation:

- Is how our bodies respond to illness, injury, germs, or chemicals.
- Is a normal process.
- Can come on suddenly and be temporary. Or it can be chronic, which means it can go on for months or years.
- Can cause symptoms like fever, rashes, swelling, or pain.
- Can happen because of ICI therapy.

TNF is a type of immune system protein that plays a role in inflammation. TNF blockers (see **Guide 2**) help prevent the protein from causing inflammation.

TNF blockers may be used to treat side effects of the:

- Lungs and heart
- Digestive system (see Chapter 5)
- Kidney (see Chapter 6)
- Muscles and joints (see Chapter 11)

Other immune blockers that may be used to treat side effects are:

- Dupilumab (Dupixent)
- Omalizumab (Xolair)
- Rituximab (Rituxan)
- Sarilumab (Kevzara)
- Tocilizumab (Actemra)

| Guide 2<br>TNF blockers |
|---|
| Adalimumab (Humira) |
| Certolizumab pegol (Cimzia) |
| Etanercept (Enbrel) |
| Golimumab (Simponi Aria, Simponi) |
| Infliximab (Remicade) |
| Vedolizumab (Entyvio) |

Immune blockers may be used to treat ICI side effects related to:

- Skin
- Lungs
- Blood and bone marrow
- Kidney and liver
- Muscles and joints

### Gabapentinoids and immunosuppressants

Gabapentinoids help with pain or itching. They may be used to treat side effects of the skin (see Chapter 3), mouth (see Chapter 4), and nerves (see Chapter 10).

Immunosuppressants like methotrexate and mycophenolate mofetil keep the immune system from attacking healthy tissue and cells by mistake. They are given orally. Methotrexate can also be given by IV (see **Guide 3**. Note: Methotrexate does not have a brand name).

- Methotrexate also helps with inflammation. It's sometimes used to treat side effects of the skin and mouth, as well as the muscles and joints (see Chapter 11).
- Mycophenolate mofetil is sometimes used to treat side effects of the:
  - Skin
  - Mouth
  - Lungs (see Chapter 8)
  - Blood and bone marrow (see Chapter 11)
  - Kidney and liver (see Chapter 6)
  - Muscles and joints

**"Immune checkpoint inhibitor therapy was a game changer for me.**

**My moments of not feeling well after infusion were minimal. From the time I entered the infusion room to the time I left, I felt supported in every way possible. My needs were always a top priority for my entire team."**

**Guide 3**
**Gabapentinoids and immunosuppressants**

Gabapentinoid
- Gabapentin (Neurontin)
- Pregabalin (Lyrica)

Mycophenolate mofetil (CellCept, Myhibbin)

## Fatigue: a common side effect

Feeling tired is common during immunotherapy. And so is fatigue. Fatigue is extreme tiredness and lack of energy. Fatigue can also be a symptom of another condition.

If you're having fatigue, your care team will order blood tests, do a physical exam, and ask about medicines you're taking. They will also test for depression, which can be related to fatigue.

Your team will assess whether fatigue is happening because of immunotherapy or if it's due to something else. You can learn about common side effects that can cause fatigue in **Guide 4**.

For more information, see *NCCN Guidelines for Patients: Fatigue and Cancer*, available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

**Guide 4**
**Side effects that can cause fatigue**

| Side effect: | To learn more about this side effect, go to: |
|---|---|
| Inflamed heart (myocarditis)<br>Inflamed lungs (pneumonitis) | *Chapter 8: Side effects of the lungs and heart* |
| Blood sugar crisis (diabetic ketoacidosis, or DKA)<br>Issues involving hormones:<br>• Hypothyroidism<br>• Thyrotoxicosis<br>• Hypophysitis | *Chapter 7: Side effects of the endocrine system* |
| Muscle weakness (myositis) | *Chapter 11: Side effects of the muscles and joints* |
| Low red blood cell count | *Chapter 12: Side effects of the blood and bone marrow* |

## Communication is key

Stay in close contact with your care team. While many side effects come on slowly, others can start and worsen quickly.

Reporting symptoms when they start can help find the cause or trigger early. It also helps prevent severe side effects. Symptoms that may seem unrelated, like diarrhea or shortness of breath, could be a sign of an ICI side effect.

Ask your care team if they would like you to report symptoms online using a patient portal, or if they prefer you to reach them another way.

Contact your care team if you have:

- Severe fatigue
- Headache
- Rash
- Cough
- Shortness of breath
- Chest pain
- Bloating
- Changes in pooping (bowel habits)
- Weight loss
- Vision changes or eye pain
- Severe muscle weakness
- Severe muscle or joint pain
- Mood changes

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**NCCN.org/patients/feedback**

## Key points

- Before you start immune checkpoint inhibitor (ICI) therapy, you'll have a physical exam and blood tests and your care team will ask about your personal and family medical history.
- It's important to carry an immunotherapy wallet card with you during treatment.
- It is recommended that people of childbearing age use birth control and not breastfeed during immunotherapy and for 5 months after the last dose.
- Inactive or killed vaccines are safe during ICI treatment.
- Many side effects of ICIs are treated with steroids.
- Fatigue is common during immunotherapy treatment, but it may be a symptom of another condition.
- It's important to tell your cancer care team if you're having side effects, even if you aren't sure they're related to your treatment.

## Questions to ask

- How should I contact you if I have a symptom or symptoms?
- Is it okay to travel during treatment with ICIs?
- Is it safe to start other medicines during ICI therapy?

## What's next?

The next chapter discusses side effects that affect the skin. Many of these side effects are common. You'll learn how to manage mild symptoms and how moderate or severe symptoms are treated.

# 3
# Side effects of the skin


20 Rash

22 Itchy skin

24 Blistering

25 Lichen planus

27 Psoriasis

28 SJS and TEN

29 Key points

29 Questions to ask

Skin issues are common during treatment with immune checkpoint inhibitors (ICIs). Most skin-related side effects are mild and can be managed without stopping immunotherapy. Very few are serious. In this chapter, you can learn about symptoms that affect the skin and how they are managed.

Symptoms that affect the skin usually start in the first several weeks of treatment. Your care team may consult with, or suggest you see, a dermatologist, a doctor who is an expert in skin diseases and conditions.

Topical steroids are typically used to manage skin-related side effects. Topical means they are put on the surface of the skin as lotions, creams, or ointments.

Topical steroids come in a variety of strengths:

- Low potency (weaker), such as hydrocortisone 0.5 or 1 percent (%)
- Moderate or high potency (stronger)
  - Moderate potency—triamcinolone 0.1%, betamethasone valerate
  - High potency—clobetasol 0.05%, fluocinonide 0.05%

Management of skin-related side effects often includes oral or IV steroid treatment.

## Rash

A skin rash has flat and bumpy patches. You may hear it called a maculopapular rash. It's often found on the upper chest, back, or both. It may spread outward. A skin rash may not feel any different than your healthy skin. But it may:

- Feel itchy
- Burn
- Feel tight

If you have a rash, your care team will do a full-body skin exam. This may include taking pictures of your skin. They will look at:

- The inside of your mouth
- Your hair and scalp
- Your fingernails and toenails

The goal is to find all areas that have a rash and learn how much of the body is affected.

Your care team:

- Will ask about symptoms and how much they are affecting your normal daily activities and self-care.
- Will ask if you've had immune-related skin issues in the past.
- May do a biopsy. A small piece of skin is removed and sent to a lab for testing.
- May order blood tests to check for other issues causing the itching, like a severe reaction syndrome.

## How rash is managed

If you have a rash, your symptoms will be treated as mild, moderate, or severe. Immunotherapy will continue, unless your symptoms are severe. A rash is:

- **Mild**, if it is only on a small part of your body
- **Moderate**, if it:
  - Is on some of your body (up to 30%)
  - Makes it hard to do your normal daily activities
- **Severe**, if it:
  - Is on much of your body (more than 30%)
  - Makes it hard for you to care for yourself

NCCN's recommendations for managing rashes are in **Guide 5**.

## Managing mild or moderate rash

Using a barrier cream or ointment is recommended. These barriers protect, soothe, and moisturize the skin.

Your care team:

- Will treat the rash with a topical steroid. Stronger topical steroids may be used for moderate symptoms.
- May treat symptoms with an oral antihistamine (allergy medicine) to control itchy skin. Oral means taken by mouth.
- If topical steroids don't help with moderate symptoms in 1 to 2 weeks, an oral steroid may be prescribed.

## Managing severe rash

Your care team will treat the rash with stronger topical steroids. An oral steroid or IV steroid will be prescribed.

A small piece of skin tissue may be removed and sent to a lab for testing. This is called a biopsy.

**Guide 5**
**Managing rash**

| | |
|---|---|
| **Mild or moderate symptoms** | • Barrier cream and topical steroid<br>• Possibly:<br>  • Oral antihistamine<br>  • Oral steroid (moderate only) |
| **Severe symptoms** | • Oral or IV steroid if the topical steroid doesn't help<br>• See a dermatologist right away<br>• Possible inpatient treatment |

## Itchy skin

If you have intense itching with or without a rash you can see, it's called pruritus. The itching may come and go. Or you may feel itchy all of the time.

If your skin is very itchy, your care team will ask if you have had inflamed skin diseases in the past. Your team will also do a full-body skin exam. They may take pictures of your skin. They will look at:

- The inside of your mouth
- Your hair and scalp
- Your fingernails and toenails

The goal is to find all areas that are very itchy. Your team may also do blood tests to learn what is causing the itching. They will look for certain antibodies (proteins that fight infection) and to see if your liver and kidneys are working as they should.

If you're itchy, your care team will treat your symptoms with an oral antihistamine. It's also important to use a fragrance-free moisturizer on your whole body.

Your symptoms will be treated further based on whether they are mild, moderate, or severe. Immunotherapy will be continued if they are mild and paused if they are moderate or severe.

**Mild** symptoms may only happen on 1 area of skin.

**Moderate itching:**

- Is very intense or covers large areas of skin.
- Comes and goes.
- Makes it hard for you to do your normal daily activities.
- Leads to skin changes as a result of scratching. This can cause bleeding, oozing, and other skin issues.

**Severe itching:**

- Is very intense or covers a large area of skin.
- Is constant.
- Makes it hard for you to care for yourself or sleep.

NCCN's recommendations for managing itching are outlined in **Guide 6**.

**Guide 6**
**Managing itchy skin**

| | |
|---|---|
| **Mild symptoms** | • Topical steroid<br>• Over-the-counter pain and itch-relief cream |
| **Severe symptoms** | • Oral or IV steroid<br>• Gabapentinoid, if needed<br>• If symptoms don't get better in a month, you may have:<br>  • A biologic<br>  • UVB light therapy |

## Managing mild itching

Your care team will treat the itchy area with a topical steroid.

You may also want to use over-the-counter pain-relief and itch-relief creams with the following ingredients:

- Hydrocortisone (0.5% or 1%) to help with pain, itching, and inflamed skin
- Diphenhydramine, to help with itching
- Pramoxine, to numb the itching, pain, or both
- Camphor, to fight infection
- Menthol, to treat pain and inflamed skin

## Managing moderate or severe itching

An oral steroid or IV steroid will be prescribed. Your care team may prescribe a gabapentinoid for itching.

If your symptoms aren't getting better in a month, your team may prescribe 1 of the following:

- A biologic medicine (dupilumab or omalizumab).
- Treatment with narrow-band ultraviolet B (UVB) light therapy. UVB light therapy uses artificial light on the skin to reduce itching.

You will see a dermatologist right away.

### UVB light therapy

**Narrow-band UVB light therapy may help improve psoriasis or lichen planus that hasn't responded to other treatments.**

## Blistering

People being treated with ICI therapy may have skin inflammation and blisters (bullous dermatitis). Blisters are sacs on the outer layer of skin that are filled with fluid.

Your care team will advise you to see a dermatologist right away. If that's not possible, your care team may perform 1 or both of these tests:

- A skin biopsy (tissue is removed and sent to a lab for testing)
- A blood test to look for certain antibodies (proteins that fight infection)

If testing or biopsy confirms bullous pemphigoid, a rare autoimmune disease of the skin, your care team may prescribe:

- A biologic medicine, rituximab or dupilumab
- IVIG with rituximab, for severe cases

Blisters can be mild, moderate, severe, or life-threatening. Symptoms are considered:

- **Mild** if they aren't causing pain. Immunotherapy may be paused. A strong topical steroid will be prescribed.
- **Moderate** if they cover 10% to 30% of your skin, are painful, and make it hard to do your normal daily routine. Immunotherapy will be paused until symptoms become mild.
- **Severe** if they cover more than 30% of your skin and make it hard to care for yourself.
- **Life-threatening** if they are on more than 30% of your skin and blood testing finds an electrolyte (certain proteins that fight infection) issue. Immunotherapy will be stopped if symptoms are severe or life-threatening.

NCCN's recommendations for managing blisters are given in **Guide 7**.

### Managing moderate blistering

An oral steroid or an IV steroid will be prescribed. If your symptoms don't get better in 3 days, your team will recommend you see a dermatologist right away. If a dermatologist is not available, your team may suggest inpatient care.

### Managing severe or life-threatening blistering

An oral steroid or IV steroid will be prescribed. You'll be treated with a biologic, IVIG, or both. Your care team will have you see a dermatologist right away.

**Guide 7**
**Managing skin blisters**

| | |
|---|---|
| **Moderate symptoms** | • Oral or IV steroid<br>• Biologic medicine<br>• If symptoms don't get better, see a dermatologist |
| **Severe or life-threatening symptoms** | • Oral or IV steroid<br>• Biologic medicine, IVIG, or both<br>• See a dermatologist right away |

Your symptoms will be treated in the hospital. If they are life-threatening, you may be treated in a hospital intensive care unit or burn unit.

## Lichen planus

Lichen planus affects the skin on the body or the inside of the mouth or vulva (in people with female sex organs).

It can cause:

- Reddish-purplish bumps and patches on the skin that are very itchy
- Sores inside of the mouth that may have white lines in a crisscross pattern
- Sores on the vulva that may have white lines in a crisscross pattern

If you have lichen planus:

- You'll have a physical exam.
- Your care team will look inside of your mouth and mucous membranes.
- You may have a skin biopsy (tissue is removed and sent to lab for testing).
- You may be asked to see a dermatologist.

Mucous membranes are the soft tissues that line the body's organs and cavities. They are found in the nose, ears, throat, genitals (sex and reproductive organs and glands), and anus.

### How lichen planus is managed

Lichen planus is treated with a strong topical steroid. Or your care team will treat your symptoms with an ointment-based topical immunosuppressant (tacrolimus). Treatments for lichen planus are listed in **Guide 8**.

Your symptoms will be treated as mild, moderate, or severe. Symptoms are considered:

- **Mild** if they cover a small part of your body, less than 10%. Immunotherapy will be continued, and your care team will check to see if your symptoms get better when treated with topical steroids.
- **Moderate** if they cover 10% to 30% of your body or don't get better when treated with topical steroids.
- **Severe** if they are on more than 30% of your body.

**Guide 8**
**Managing lichen planus**

| | |
|---|---|
| **Moderate symptoms** | • Oral antihistamine<br>• Oral steroid<br>• UVB light therapy if recommended<br>• Symptoms will be treated as severe if they don't get better<br>• You may see a dermatologist |
| **Severe symptoms** | • Oral or IV steroid<br>• You may see a dermatologist<br>• If needed:<br>  • An immunosuppressant<br>  • Acitretin<br>  • Dupilumab |

If symptoms are moderate or severe, your immunotherapy will be paused.

### Topical steroids for lichen planus

The form of the topical steroid depends on where symptoms are on your body. If they affect:

- Your mucous membranes, a gel-based steroid may be used
- Your scalp, a solution-based (lotion) steroid may be used
- Other areas of the body, a cream, lotion, or ointment may be used

### Managing moderate lichen planus

Your care team will try an oral antihistamine to help with itching. Your care team will prescribe an oral steroid.

You may begin treatment with narrow-band UVB light therapy. UVB light therapy uses artificial light on the skin to reduce itching.

If your symptoms don't get better after these treatments, they will be treated as severe.

### Managing severe lichen planus

An oral steroid or an IV steroid will be prescribed. Your care team may:

- Suggest you see a dermatologist
- Prescribe immunosuppressant medicines, including:
  - Azathioprine (Imuran)
  - Cyclosporine
  - Hydroxychloroquine (Plaquenil, Sovuna)
  - Methotrexate
  - Mycophenolate mofetil
- Prescribe acitretin (a retinoid) if there is no chance you could become pregnant
- Prescribe dupilumab

**What are mucous membranes?**

Also called mucosa, they are the soft, moist lining of the mouth, nose, and other parts of the body.

## Psoriasis

Psoriasis is patches of thick, red, and scaly skin. These patches often show up:

- Around the joints (elbows, knees, shins)
- On the scalp and behind the ears
- Around the navel (belly button)

If you have symptoms of psoriasis, your care team will perform a physical exam. They may check your oral and dental health. They may also perform a skin biopsy (tissue is removed and sent to a lab for testing). They may discuss your symptoms with, or ask you to see, a dermatologist.

### How psoriasis is managed

Your symptoms will be treated as mild, moderate, or severe. Symptoms are considered:

- **Mild** if they cover a small part of your body (less than 10%).
- **Moderate** if they cover 10% to 30% of your body or if they don't get better when treated with strong topical steroids. Topical means the steroid is put on the skin.
- **Severe** if they are on more than 30% of your body.

If psoriasis is moderate or severe, your immunotherapy will be paused.

NCCN's recommendations for managing psoriasis are outlined in **Guide 9**.

### Managing mild psoriasis

Your care team will treat your symptoms with a strong topical steroid. Other topical medicines, such as tacrolimus (an immunosuppressant) or calcipotriene (a kind of vitamin D), may be used in skin folds or the face for long-term management.

### Managing moderate and severe psoriasis

You may begin treatment with narrow-band UVB light therapy. UVB light therapy uses artificial light on the skin to reduce itching.

Your care team may also prescribe apremilast (Otezla) or acitretin. These drugs target the inflammation. If your symptoms don't get better after the recommended treatments, symptoms will be treated as severe.

**Guide 9**
**Managing psoriasis**

| | |
|---|---|
| **Mild symptoms** | • Topical steroid<br>• Other topical medicines |
| **Severe symptoms** | • Topical steroid<br>• Other topical medicines<br>• UVB light therapy<br>• If needed:<br>  • Acitretin (if no chance for pregnancy)<br>  • Apremilast<br>  • Cyclosporine (only for severe symptoms)<br>  • Methotrexate (only for severe symptoms)<br>• See a dermatologist |

Your care team will recommend you see a dermatologist to discuss possible treatment with other biologics that treat inflammation.

For severe symptoms, your team may also prescribe cyclosporine or methotrexate.

## SJS and TEN

Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) are very serious skin and mucous membrane conditions caused by allergic reactions to certain medicines.

SJS and TEN can cause blistering. The blisters separate from the skin or mucous membrane and peel off.

Your care team will look at any medicine (such as an antibiotic or anticonvulsant) you're taking to see if they are the cause of these conditions.

### How SJS or TEN is managed

Your care team will advise you to see a dermatologist right away. If that's not possible, your care team may perform a skin biopsy (tissue is removed and sent to a lab for testing).

You'll also see an ophthalmologist (an expert in vision and eye disease) and urologist (an expert in urinary tract conditions) because SJS and TEN can affect the eyes, the sex and reproductive organs, and the urinary tract.

Your immunotherapy will be stopped. Your care team will prescribe an oral steroid or IV steroid.

You may also be given IVIG. Other immunosuppressant therapies (etanercept or cyclosporine) may be prescribed. You'll be cared for as an inpatient, in the hospital.

**SJS and TEN are rare side effects of the skin that can also affect the eyes, sex and reproductive organs, and urinary tract.**

## Key points

- Most skin-related side effects of immunotherapy are mild and can be managed without stopping immune checkpoint inhibitor (ICI) treatment.
- Common side effects of ICI treatment include rash, itching, and blisters.
- Topical steroids are used to treat many skin-related side effects. Topical means the medicine is put on the skin.
- If you have a skin-related side effect, you'll likely see a dermatologist, a doctor who's an expert in skin diseases and conditions.
- You may have a skin biopsy. This is when a very small piece of skin is sent to a lab for testing.
- Symptoms of skin-related side effects may be managed with oral (given by mouth) or IV (given by needle into a vein through the skin) steroids.
- Rare, serious side effects of ICI treatment include bullous pemphigoid, a rare autoimmune skin disorder, and Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis (TEN).

## Questions to ask

- Should I talk to my care team if my skin is itchy but I have no other symptoms?
- What is the best soap or cleanser to use if I have a rash or itchy skin?
- If I had eczema, psoriasis, or another skin condition in the past, should I tell my care team, even if it's under control?
- Will the dermatologist have experience treating people who are undergoing immunotherapy for cancer treatment?
- How will I know if I should go to the hospital for a side effect?

## What's next?

In the next chapter, you'll learn about ICI side effects of the mouth, tests you may have, and how your side effects are managed.

# 4
# Side effects of the mouth

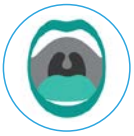

31 Inflamed lining of the mouth

33 Burning mouth or mouth pain

34 Dry mouth syndrome

36 Key points

36 Questions to ask



In this chapter, you'll learn how the mouth and its lining (oral mucosa) can be affected by treatment with immune checkpoint inhibitors (ICIs). You'll learn how these conditions are managed, and if and how they will affect your immunotherapy treatment.

If you have any mouth-related symptoms, like burning, pain, or dryness due to immunotherapy treatment, your care team will recommend eating foods that don't make your symptoms feel worse. You'll be asked to avoid foods (and drinks) that are crunchy, spicy, and acidic (like citrus fruits and juices, and coffee).

Your team will likely recommend that you see a dentist if you have symptoms that affect your mouth. A dentist can help keep your mouth clean and help prevent cavities.

## Inflamed lining of the mouth

If the lining of your mouth is inflamed, it's important to take good care of your teeth and mouth.

Your care team will:

- Examine the lining for inflammation—irritated gums or throat, mouth sores, and lichen planus. (See *Chapter 3: Side effects of the skin* to learn about lichen planus and how it's treated.)
- Ask about pain and if it's preventing you from eating as you normally do.
- Recommend an oral rinse that contains chlorhexidine. This medicine kills or prevents bacteria from growing in your mouth. It also reduces swelling and irritation of the gums.
- Order blood tests if you have mouth sores. These tests look for certain antibodies (proteins that fight infection) that confirm a serious blistering condition that may be

**There are things you can do at home to help manage mild side effects that affect your mouth.**

**A rinse with chlorhexidine helps reduce swelling and irritated gums.**

caused by immunotherapy (see *Chapter 3: Side effects of the skin*).

- Check if your symptoms are due to other therapies you're receiving.

Your care team may remove a very small piece of tissue from your mouth for a biopsy. This means the tissue will be sent to a lab to test for certain fungal or viral infections.

Your care team may evaluate your nutrition and give you supplements, if necessary.

## Medicine that may be used

Topical medicine may be prescribed for the skin or lining of the mouth to treat inflammation. These include:

- Liquid-based steroids – dexamethasone or budesonide (Entocort, EC, Uceris)
- Gel-based steroids – fluocinonide, clobetasol, or betamethasone dipropionate
- Ointment-based, non-steroidal immunosuppressants – tacrolimus

If symptoms get worse or don't respond after a month of treatment, your team may prescribe methotrexate or mycophenolate mofetil.

## Other specialists you may see

If available, your health care team may refer you to:

- A dermatologist, a doctor who's an expert in skin issues and diseases.
- An oral pathologist, a dentist who's an expert in diagnosing and treating mouth conditions.
- An ENT (ear, nose, and throat) doctor to help manage inflamed mouth that doesn't go away, or if the back of the mouth, upper throat, or both are affected.

## Managing symptoms of inflamed mouth

Symptoms of inflamed mouth can be:

- **Mild**, if only some foods are too difficult to eat. Your immunotherapy will be continued.
- **Moderate**, if you can only eat soft or mashed foods. Your immunotherapy will be paused.
- **Severe**, if you're unable to eat at all. Your immunotherapy will be paused.

NCCN's recommendations for managing inflamed mouth are highlighted in **Guide 10**.

**Guide 10**
**Managing inflamed mouth**

| | |
|---|---|
| **Mild symptoms** | • Topical pain reliever<br>• If needed:<br>  • Medicine to reduce stomach acid<br>  • Steroid injected into mouth sores |
| **Moderate or severe symptoms** | • Oral or IV steroid<br>• If needed, biopsy of your lip or mouth<br>• If moderate, your team may inject a steroid into mouth sores<br>• If severe, you'll be treated in the hospital |

### Managing mild symptoms of inflamed mouth

A proton pump inhibitor or histamine-2 blocker may be used to help reduce stomach acid that can irritate the throat.

You can make a topical pain reliever at home called magic mouthwash with equal parts:

- Antihistamine (diphenhydramine) for itch relief (children's liquid Benadryl)
- Antacid (Maalox or Mylanta)
- Viscous lidocaine to numb the mouth (available through a prescription)

If your care team has treated your mouth with a topical steroid and the sores don't get better, a steroid may be injected with a needle into the sores.

### Managing moderate or severe symptoms of inflamed mouth

You'll be given an oral or IV steroid for moderate or severe symptoms. You'll have a biopsy of your lip or mouth if your team thinks it will be useful and it hasn't been done yet.

If your symptoms are moderate and sores in your mouth don't get better after treatment with a topical steroid, your team may inject a steroid into the sores with a needle.

If symptoms are severe, you'll be treated in the hospital.

## Burning mouth or mouth pain

Having pain or feeling like your mouth is burning can be a side effect of treatment with ICIs (**Guide 11**). This is called oral dysesthesia. People often report that the pain seems worse than it should given other symptoms.

It's best to avoid foods (and drinks) that are crunchy, spicy, acidic (like citrus fruits and juices, and coffee).

Symptoms of burning mouth and mouth pain can be:

- **Mild**, if they don't affect eating and drinking. Immunotherapy will be continued.
- **Moderate**, if they do affect eating and drinking.
- **Severe**, if they are very painful and tube- or IV-feeding is needed.

| Guide 11<br>Burning mouth symptoms |
| --- |
| Changes of the skin inside the mouth |
| Mouth sensitivity |
| Altered sense of taste (food tastes sour, sweet, metallic, or bitter) |
| Unpleasant taste in the mouth |
| Changes in the way the mouth feels |

Your immunotherapy will be paused if symptoms are moderate or severe.

Your care team will:

- Rule out other possible causes of the burning or pain
- Check your jaw and teeth for infection or abnormal growth
- Look at your tongue and the skin inside of your mouth
- Ask you how much the symptoms are affecting your ability to eat and drink

### Managing burning mouth symptoms

NCCN's recommendations for managing burning mouth or mouth pain are given in **Guide 12**.

If a topical steroid is used, it will likely be a gel or rinse that you swish and hold in your mouth, then spit out. Viscous lidocaine is a rinse that numbs the mouth. It helps relieve pain and swelling. Gabapentin is another kind of oral pain reliever that may be used.

**Guide 12**
**Managing burning mouth or mouth pain**

| | |
|---|---|
| **Mild symptoms** | Topical steroid or viscous lidocaine |
| **Moderate or severe symptoms** | 1 of the following, if needed:<br>• Strong topical steroid<br>• Viscous lidocaine<br>• Gabapentin |

## Dry mouth syndrome

If your mouth is extremely dry (sicca syndrome), your care team will:

- Ask if you're taking any medicines that are making your mouth dry. If so, you'll be asked to stop taking them if possible.
- Check to make sure your body has enough water to work well.
- Check the thickness of your saliva and if it's "pooling" in 1 spot in your mouth.
- Look at the health of your gums and teeth.
- Test your blood for certain proteins.
- Treat your symptoms as mild, moderate, or severe.

Symptoms are mild if your saliva is dry or thick but you're able to eat the foods you usually do. Your immunotherapy will be continued.

Symptoms are moderate or severe if you:

- Need to drink a lot to swallow dry food
- Can only eat soft, moist, or mashed food or can't eat at all
- Need an oral lubricant that keeps your mouth moist

If your symptoms are moderate or severe, your immunotherapy will be paused.

### Managing dry mouth symptoms

Ways to manage dry mouth symptoms are listed in **Guide 13**. You'll also be asked to avoid foods (and drinks) that are crunchy, spicy, and acidic (like citrus fruits and juices, and coffee).

Topical comfort treatments are:

- Sipping water
- Using saliva substitutes
- Using mouth rinses or sprays with no alcohol, or children's toothpaste

### Managing mild dry mouth

Products that help make more saliva are recommended. This includes chewing sugarless gum and sucking on lozenges or candies.

### Managing moderate or severe dry mouth

An oral steroid will be used for 2 to 4 weeks. Your care team will adjust the dose depending on whether your symptoms respond to the steroid. Other oral medicines to help the mouth produce more saliva (cevimeline or pilocarpine) will be used.

If your symptoms are severe, your care team may want you to be treated in the hospital (inpatient).

### Other specialists you may see

If you have moderate or severe symptoms, your care team will recommend you see a rheumatologist. This is a doctor who is an expert in conditions of the bones, joints, muscles, and tendons, as well as autoimmune diseases.

Dry mouth can lead to tooth decay and tooth loss. Your team will want you to see an oral pathologist, if available. This is a dentist who is an expert in diagnosing and treating mouth conditions.

**Let us know what you think!**

**Please take a moment to complete an online survey about the NCCN Guidelines for Patients.**
**NCCN.org/patients/response**

| Guide 13<br>Managing dry mouth | |
|---|---|
| **Mild symptoms** | • Topical comfort<br>• Products that help make more saliva |
| **Moderate or severe symptoms** | • Oral steroid<br>• Other oral medicines<br>• Your care team will refer you to:<br>  • A rheumatologist<br>  • An oral pathologist<br>• If needed,<br>  • Inpatient care<br>  • More topical treatments |

## Key points

- Side effects of the mouth are inflammation, burning or pain, and dry mouth.
- If you have a mouth-related side effect, it's important to see a dentist who can help keep your mouth clean and prevent cavities.
- You'll be asked to avoid foods (and drinks) that are crunchy, spicy, and acidic (like citrus fruits and juices, and coffee).
- Your immunotherapy can be continued if your mouth-related side effects are mild.
- Steroids are usually the first treatment for moderate to severe mouth symptoms.
- Other medicines or treatments may be used to manage specific symptoms like swelling or dryness.

## Questions to ask

- Should I see my dentist before starting treatment with ICIs?
- Is it ok to have dental procedures done during ICI treatment?
- What kinds of food should I eat?
- What kinds of food or drinks should I avoid?
- Why is a biopsy being done?

## What's next?

In Chapter 5, side effects of the digestive system, or gastrointestinal (GI) tract, will be discussed. The GI tract is made up of the upper and lower digestive systems. Side effects of the GI tract are common during ICI treatment.

# 5
# Side effects of the digestive system


38 What is the digestive system?

39 Upper and lower digestive tracts

40 Lower GI symptoms

43 Upper GI symptoms

45 Key points

45 Questions to ask

This chapter explains immunotherapy side effects of the digestive system. This is also called the gastrointestinal (GI) tract. GI symptoms are common during treatment with immune checkpoint inhibitors (ICIs).

You'll learn how to manage mild symptoms. And you'll find out what to expect if your symptoms are more severe and making it hard to care for yourself.

Be sure to let your cancer care team know if you're having digestive system side effects. This will enable them to help you manage symptoms and to make decisions about your treatment and care.

## What is the digestive system?

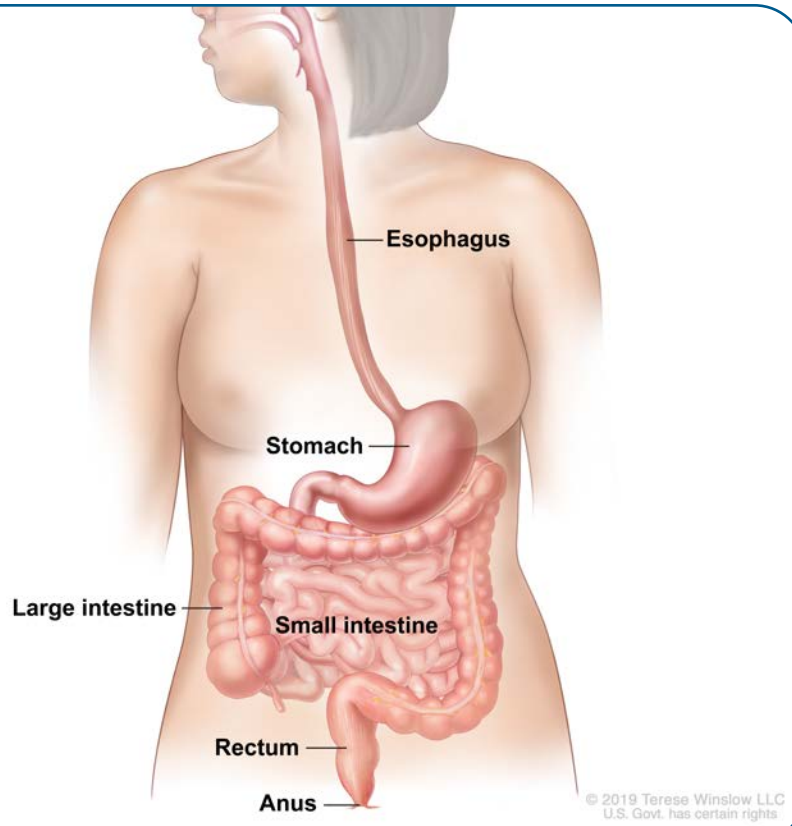
The digestive system is made up of the organs that take in what we eat and drink. Food and liquid are broken down into the materials the body uses for growth, energy, and tissue repair. What the body cannot use leaves as a bowel movement (poop or stool).

The digestive system includes the:

- Esophagus (swallowing tube)
- Stomach
- First part of the small intestine (duodenum)
- Large intestine (colon)
- Rectum
- Anus

### The digestive system

**The food we eat passes through the esophagus into the stomach. It gets broken down into a liquid and enters the small intestine. The large intestine prepares food to leave the body as a bowel movement (poop, or stool).**



### Know your baseline bowel habits

All of us have different bowel habits. Our usual habits are called baseline (starting before immunotherapy) bowel habits. It's good to know yours so you can tell if you have bowel changes when you start treatment with immune checkpoint inhibitors (ICIs).

Knowing how often you usually have a bowel movement is helpful.

It also helps to know the usual consistency of your bowel movements. Consistency is how thick or firm a bowel movement is.

Your care team will decide how to manage your side effects based on whether you're having diarrhea or other symptoms of inflamed bowel. They will also ask:

- How many bowel movements you are having above your baseline
- If the symptoms are making it hard to perform daily tasks, like showering, dressing, or walking

## Upper and lower digestive tracts

The gastrointestinal (GI) system has an upper tract. This is where digestion begins when you eat food. It includes the esophagus, stomach, and the part of the small intestine called the duodenum. Some side effects affect this part of the GI system.

Some side effects affect the lower digestive tract. This includes the large intestine (colon), rectum, and anus.

Common GI symptoms include diarrhea and belly pain. Diarrhea is pooping more often but can also be described as having watery or loose stools. Diarrhea and belly pain can be symptoms in both the upper and lower digestive tracts.

Other GI side effects are listed in **Guide 14**.

| **Guide 14**<br>**Lower GI tract (including inflamed bowel) symptoms** |
|---|
| Loose and frequent poops (diarrhea) |
| Cramps |
| Feeling the urge to poop |
| Fever |
| Blood and mucus in stool |
| A need to poop at night that wakes you up |

## Lower GI symptoms

In the lower GI tract, ICIs can cause diarrhea and inflamed bowel symptoms. Other symptoms of inflamed bowel are listed in **Guide 14**.

Here are some other things to know about lower GI side effects:

- These symptoms are common in people being treated with ICIs.
- Symptoms often start in the first 6 to 8 weeks of ICI treatment.
- The large intestine is also called the colon. Inflammation of the large intestine is called colitis.

You can learn how your team decides if symptoms are mild, moderate, or severe in **Guide 15**. This is important because your care team will make decisions about how to manage your symptoms based on these groupings.

Severe symptoms may also include:

- Abnormal or unstable blood pressure
- Serious or life-threatening colitis

If your symptoms are severe, you may be admitted to the hospital.

Keeping track of your symptoms and talking to your care team will help your team manage your symptoms and decide if other tests are needed.

It's important to be aware of:

- How often you poop
- The consistency of your poop
- If there is blood in your poop
- If you are pooping overnight
- Changes in your weight

**Guide 15**
**Mild, moderate, and severe lower GI symptoms**

| | |
|---|---|
| **Mild** | • 3 bowel movements or fewer per day than your baseline<br>• No symptoms of inflamed bowel |
| **Moderate** | • 4 to 6 more bowel movements per day than your baseline<br>• Inflamed bowel symptoms<br>• Symptoms don't affect your ability to care for yourself |
| **Severe** | • 7 or more bowel movements per day than your baseline<br>• Inflamed bowel symptoms<br>• Symptoms affect your ability to care for yourself |

## Tests you may have for lower GI symptoms

If you're having diarrhea and inflamed bowel symptoms, your care team may test your stool. For moderate or severe symptoms, your team may suggest you see a GI specialist and have other tests.

**Stool testing** is when your health care team takes a sample of your poop and sends it to a lab. They will do this to see if your symptoms are caused by something other than immunotherapy. Your stool may be tested for:

- Bacteria, viruses, and parasites
- Certain proteins that could mean your large bowel is inflamed

For more severe symptoms, other testing may include:

- A CT of the belly and pelvis
- Procedures to see inside of the bowel, esophagus, and stomach

You can read more about these tests and procedures later in this chapter.

## Managing lower GI symptoms

There are things you can do at home to help manage mild lower GI symptoms. These include:

- Drink more fluids than usual.
- Eat low-fiber, lactose-free food.
- Try a BRAT (bananas, rice, applesauce, toast) diet.
- Take an anti-diarrheal medicine like loperamide (Imodium) or diphenoxylate/atropine (Lomotil). You can get these medicines without a prescription.

Your care team may pause your immunotherapy if you have mild lower GI symptoms. They will keep track of your symptoms and decide what to do next based on your symptoms. They may suggest taking an anti-diarrheal medicine.

If mild lower GI symptoms stay the same or get worse after 2 or 3 days of taking anti-diarrheal medicine, your care team may want you to:

- Have lab tests to check for infection, if you haven't had them yet
- Stop taking anti-diarrheal medicine to see if your symptoms are a response to immunotherapy, or if they are being caused by something else

If you continue to have symptoms or your symptoms get worse, your care team will test your stool for certain proteins that may be causing the symptoms. If the stool test shows that you:

- **Have these proteins**, your care team will treat you for moderate symptoms
- **Do not have these proteins** or an infection, your care team will keep managing your symptoms as mild. They may prescribe other medicines that treat digestion issues (mesalamine and cholestyramine)

## Managing moderate lower GI symptoms

If your symptoms are moderate, your care team will pause your immunotherapy. Your team may recommend you see a GI expert.

Symptoms may be treated with budesonide, an oral steroid, or you may be given an IV steroid.

- If your symptoms aren't getting better in 3 days, you may be treated with an IV steroid.
  - If your symptoms aren't getting better and test results show your bowel is very inflamed or there is a different bowel condition present, your care team may treat your symptoms with biologics (infliximab or vedolizumab).
  - You may also be given immunosuppressive treatment (tofacitinib or ustekinumab).

## Managing severe diarrhea and inflamed bowel

If you have severe symptoms, your care team will stop some or all immunotherapy treatment. You may be admitted to the hospital. Your team will recommend you see a GI expert.

Your symptoms will be treated with an IV steroid, and then an oral steroid.

- If your symptoms don't get better in a day or 2 and test results show your bowel is very inflamed or there is a different bowel condition present, your care team will continue to treat your symptoms with steroids and may use biologic therapy (infliximab or vedolizumab).
  - You may also be given immunosuppressive treatment (tofacitinib or ustekinumab).

If you continue to have diarrhea that doesn't get better after the treatments discussed here, your team may test for other digestive or pancreatic issues.

If immunosuppressive treatment doesn't help moderate or severe inflamed bowel, you may have a procedure to add healthy stool from a donor into your colon.

## Tests you may have for moderate or severe GI symptoms

You may have procedures or tests if your symptoms are moderate or severe, including:

- **Colonoscopy** to see and remove any abnormal tissue from the colon. A thin device is inserted through the anus, up the rectum, and into the colon. The device has a light, camera, and cutting tool.
- **CT scan with contrast** to see inside of the abdomen (belly) and pelvis. Contrast is used to make the pictures clearer. Contrast leaves the body in your urine right after the test.
- **Flexible sigmoidoscopy** to see and remove any abnormal tissue from the rectum and the last part of the colon. A thin device is inserted through the anus into the rectum. The device has a light, camera, and cutting tool.
- **Upper GI endoscopy** to see inside of the esophagus, stomach, and upper parts of the small intestine.
  - **A biopsy** may be performed during an upper GI endoscopy. A biopsy is a procedure that removes tissue or fluid used for diagnosing diseases and conditions. If your stool has certain proteins in it, you may have further endoscopies during treatment.

## Upper GI symptoms

Some side effects affect the upper digestive tract, including the esophagus, stomach, and some of the small intestine. Possible upper GI symptoms are listed in **Guide 16**.

If you're having upper GI side effects, your care team will ask about possible risk factors that may be causing the symptoms. These risk factors include other medicines or cancer therapies, or you may have celiac disease.

You may also have an upper GI endoscopy, which is explained earlier in this chapter. A biopsy may be performed during this procedure.

You can learn how your team decides if upper GI symptoms are mild, moderate, or severe in **Guide 17**. Your care team will decide how to manage your symptoms based on these groupings.

| Guide 16<br>Upper GI tract symptoms |
|---|
| Nausea |
| Vomiting |
| Difficulty, pain with swallowing, or both |
| Reflux (heartburn) |
| Vomiting blood |
| Indigestion |
| Black poop |
| Lack of interest in eating |
| Feeling full after eating too little |
| Bloating |
| Weight loss |

### Managing mild upper GI symptoms

If symptoms are mild, immunotherapy will be continued.

Things you can do to help manage your lower GI symptoms include:

- Drink more liquids than usual.
- Change your diet to help with symptoms.
- Take medicine to treat nausea.

| Guide 17<br>Mild and moderate to severe upper GI symptoms | |
|---|---|
| **Symptoms are mild if you are:** | • Able to eat a regular diet<br>• Hydrated (your body has enough fluid)<br>• Not having weight changes |
| **Symptoms are moderate to severe if you are:** | • Not able to eat a regular diet<br>• Dehydrated (your body doesn't have enough fluid)<br>• Losing weight<br>• Having abnormal or unstable blood pressure |

Talk to your care team about other things you can do to help reduce symptoms.

You may be given medicine used to:

- Treat ulcers (sucralfate)
- Reduce stomach acid (proton pump inhibitors)
- Treat pain (acetaminophen or aspirin)

## Managing moderate to severe upper GI symptoms

Your immunotherapy may be paused if your symptoms are severe.

You may have an upper endoscopy to test for infections or inflammation that might be causing symptoms. You may also have tests to see how well food is moving through your digestive tract.

Your team:

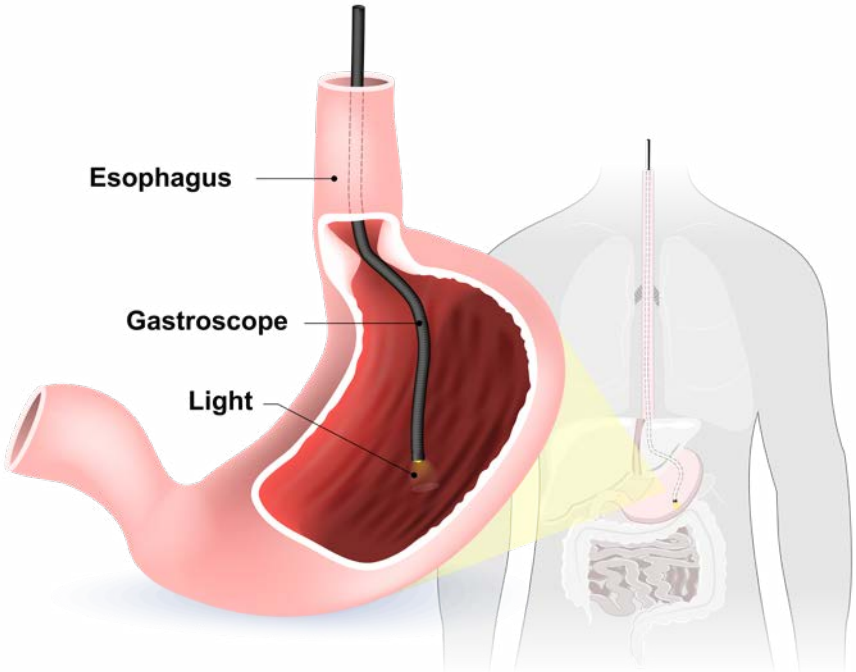
- May want to treat your symptoms in the hospital.
- Will refer you to a GI specialist.
- Will prescribe an oral and IV steroid.

If steroids don't help symptoms, your care team may treat them with biologic therapy (infliximab or vedolizumab).

If you continue to have diarrhea that doesn't get better after the treatments described, your team may test for other conditions. These can include pancreatic exocrine insufficiency and celiac disease.

### Upper gastrointestinal (GI) endoscopy

**In an upper GI endoscopy, a device (gastroscope) is guided down the throat into the esophagus, stomach, and upper parts of the small intestine.**



## Key points

- Diarrhea and inflamed bowel symptoms are common lower gastrointestinal (GI) side effects of immune checkpoint inhibitors (ICIs).
- Diarrhea and inflamed bowel symptoms often start in the first 6 to 8 weeks of immunotherapy treatment.
- Upper GI symptoms can include nausea, vomiting, cramping, and pain.
- Drinking more fluids and eating low-fiber, lactose-free foods can help reduce mild diarrhea and colitis symptoms.
- Keep track of how often you usually poop and your poop's consistency, so you know what's normal for you before you start ICI treatment.
- If your symptoms of diarrhea and inflamed bowel are moderate or severe, you may have tests that look inside of your body. These will provide more information about what is causing symptoms.
- Moderate and severe symptoms of the digestive tract are often treated with steroids.

## Questions to ask

- When should I talk to my care team if I'm having diarrhea?
- Are there any foods I should avoid during treatment?
- Is it safe for me to take ibuprofen for pain relief?
- Should I see a GI specialist before I start treatment?
- What should I do if I can't poop or if it hurts to poop?

## What's next?

The next chapter discusses side effects of the kidneys and liver. These side effects do not always cause symptoms. But they can be serious.

# 6

# Side effects of the kidneys and liver


47 Kidney side effects

48 Liver side effects

50 Key points

50 Questions to ask

In this chapter, you'll read about side effects of immune checkpoint inhibitors (ICIs) that can affect the kidneys and the liver.

- **The kidneys** are located below the ribcage behind your belly. They remove waste and extra water from the blood.
- **The liver** is located on the upper right side of the belly, protected by the ribcage. The liver has many jobs. It helps filter the blood. It also helps the body digest food.

Side effects of these organs may not cause symptoms. But when side effects do occur, they can become very severe.

## Kidney side effects

### High creatinine level

The kidneys filter a waste product called creatinine. The level of creatinine in your blood and protein in your urine (pee) will be checked during immunotherapy. A high level of creatinine signals that the kidneys aren't working well or are in crisis.

Your team will review all medicines you take, including non-steroidal anti-inflammatory drugs and proton pump inhibitors (PPIs), and order urine and imaging tests. Urine may be collected over the course of a full day and tested. Tests are done to confirm that a high creatinine level is caused by immunotherapy and not something else, like an infection or kidney stones.

### Acute kidney injury

Acute kidney injury (AKI) happens when the kidneys suddenly stop filtering blood. This kidney crisis comes on quickly, usually in less than a few days.

A nephrologist, a doctor who's a kidney expert, will likely be involved in your care.

Your team:

- Will adjust the dose or stop medicines that may be affecting creatinine levels
- Will prescribe an oral or IV steroid with a histamine blocker (not a PPI) to manage stomach acid
- May increase oral or IV fluids and retest
- Will check creatinine every 3 to 7 days

AKI is grouped in stages based on how high creatinine levels are. Stages are represented by numbers 1 to 3. The higher the number, the more severe the AKI.

### Managing stage 1 AKI

For stage 1 AKI, your team may pause your immunotherapy.

### Managing stage 2 AKI

Immunotherapy will be paused. You'll take an oral steroid until creatinine levels get better. Then the steroid will be tapered over 4 to 6 weeks.

If creatinine levels don't improve after 5 to 7 days of steroid treatment or a new protein is found in the urine, you may have a kidney biopsy. If levels don't begin to improve after 1 week, an IV steroid will be added.

### Managing stage 3 AKI

Stage 3 AKI means creatinine levels are very high, and kidney therapy is needed. You may be admitted to the hospital.

You'll take an oral and an IV steroid until creatinine levels get better. Then steroids will be tapered over 4 to 6 weeks.

If creatinine levels don't begin to improve after 5 to 7 days of steroids or a new protein is found in the urine, you'll have a kidney biopsy. Based on the biopsy results and creatinine levels after 4 to 6 weeks of steroids, you may be given 1 of the following:

- Azathioprine
- Infliximab
- Mycophenolate mofetil
- Rituximab

## Liver side effects

Treatment with ICIs can harm the liver. But there are often no symptoms. Liver injury isn't usually serious, but it can become severe.

Liver function blood tests measure levels of **bilirubin** and certain **enzymes**.

- **Bilirubin** is a waste product found in bile, which helps the digestive system break down food.
- **Enzymes** are proteins that help the liver do some of its jobs.

High levels of bilirubin or enzymes may be a sign of liver disease. Your immunotherapy will be paused or stopped if high levels are found. Your team will:

- Make sure something else isn't causing liver issues. Other causes include viruses, alcohol use, and medicines and supplements you may be taking.
- Order blood tests to make sure high liver function levels aren't due other conditions, such as:
  - Myositis (muscle weakness)
  - Myocarditis (inflamed heart)
  - Guide 13-like syndrome (a rare, serious problem of the immune system)
- Order imaging tests.
- Prescribe a steroid and ursodiol, a medicine used to treat liver diseases.

Your team may:

- Test your levels of bilirubin, enzymes, and liver protein on a regular basis.
- Monitor your blood's ability to clot.
- Suggest you see a gastroenterologist, a doctor who's an expert in issues that affect the digestive system. *(See Chapter 5: Side effects of the digestive system.)*
- Treat you in the hospital.

### Managing serious liver-related side effects

If levels of enzymes and bilirubin get very high, risk for liver failure increases. This is a medical emergency.

Your team will involve a gastroenterologist and a hepatologist in your care. A hepatologist is a doctor who's an expert in liver issues.

You'll have frequent blood tests to learn how well your blood is clotting. You may also have a liver biopsy.

If you're at risk for liver failure, your care team will start you on an oral steroid or IV steroid. When liver tests show improvement, the steroid will be tapered.

If steroids don't help after a day or 2, you'll be treated with 1 of these immunosuppressants:

- Mycophenolate mofetil
- Tacrolimus

Mycophenolate mofetil may also be given at the same time as steroids.

If those don't help, tocilizumab or other immunosuppressant therapy (azathioprine or anti-thymocyte globulin) may be started. Tocilizumab is used with caution in people with a history of certain colon issues.

You'll see a gastroenterologist and hepatologist in a week or 2 if either:

- Enzyme and bilirubin levels haven't improved
- 2 immunosuppressive treatments haven't helped

You may be admitted to the hospital (inpatient care), especially if:

- Your bilirubin level is very high
- Your blood isn't clotting well
- There's fluid build-up in your belly
- Your brain isn't working properly

**You can learn more about:**

**Myocarditis in *Chapter 8: Side effects of the lungs and heart***

**Myositis in *Chapter 11: Side effects of the muscles and joints***

**HLH-like syndrome in *Chapter 12: Side effects of the blood and bone marrow***

Your liver will be tested every 1 to 3 days.

## Key points

- Treatment with immune checkpoint inhibitors (ICIs) can affect how well the kidneys filter out waste products.
- Acute kidney injury (AKI) happens when the kidneys stop filtering blood.
- If AKI is very serious, you will be treated with a steroid. You may need to be treated in the hospital.
- Liver issues don't always cause symptoms, but conditions of the liver can become severe.
- High levels of bilirubin (found in bile) and certain enzymes (proteins) can be a sign of liver disease.
- If levels of bilirubin and certain enzymes in the liver get very high, risk for liver failure increases.
- If you're at risk for liver failure, you'll be treated with a steroid and possibly other medicines.

## Questions to ask

- Why am I having a kidney biopsy?
- Why am I being given steroids and other medicines if I'm not having symptoms?
- What does it mean if my blood isn't clotting well?
- I missed a steroid dose. What should I do?

## What's next?

In the next chapter, side effects of the pancreas and pituitary and thyroid glands will be discussed. They are part of the endocrine system, which makes hormones. You'll learn about symptoms and how they are managed.

# 7

# Side effects of the endocrine system


52 Side effects of the pancreas

55 Side effects of the pituitary and thyroid glands

57 Key points

57 Questions to ask

Treatment with immune checkpoint inhibitors (ICIs) can affect hormones, a key product of the endocrine system. Hormones are chemicals our bodies make that help manage the way they work. The endocrine system, is made up of glands, organs, and tissues.

People who develop issues that affect the hormones will likely have their care managed by an endocrinologist, a doctor who's an expert in those conditions.

In this chapter, you can read about how the organs and glands that relate to hormones can be affected by immune checkpoint inhibitor (ICI) treatment. You can also read about how these side effects are managed.

## Side effects of the pancreas

The pancreas is a gland behind the stomach. It is part of both the endocrine system and the digestive system. The pancreas makes:

- Insulin, a hormone that helps control blood sugar. Insulin will be discussed later in this chapter.
- Lipase, an enzyme that helps our bodies digest fats.

Levels of lipase are measured in blood tests. They can become high during ICI treatment. Having high lipase levels may not cause symptoms. High levels can be a sign of pancreas damage, called pancreatitis.

Even if lipase levels are high, you may be able to continue immunotherapy if you do not have pancreatitis. Pancreatitis can cause symptoms. They are listed in **Guide 18**.

High lipase levels can also be caused by 1 or more of the following:

- A blockage in the bowel (large intestine) or other bowel issues
- An issue with the stomach muscles
- Nausea and vomiting
- Medicine you are taking
- Alcohol use
- Diabetes mellitus

If your lipase levels become very high or your care team thinks you may have pancreatitis,

| **Guide 18**<br>**Pancreatitis symptoms** |
|---|
| Nausea |
| Bloating |
| Belching |
| Belly or back pain |
| Vomiting |
| Abnormal heart rate (life-threatening) |

your immunotherapy may be paused or stopped.

## High lipase levels and pancreatitis

You'll have a CT with contrast to see if the pancreas is inflamed. If your team thinks you have pancreatitis because of your high lipase levels but the CT does not confirm it, you may also have an MRI of the pancreas and bile ducts.

Your team will test for other causes of high lipase levels. These include:

- Alcohol use or issues of the liver, gallbladder, or bile ducts
- Other issues of the pancreas
- Digestive issues, including diarrhea and colitis (discussed in *Chapter 5: Sides effect of the digestive system).* If a digestive condition is found, the side effect will be managed as a chronic (persistent) condition.

If an issue other than pancreatitis is found, your team will treat that condition. If they suspect pancreatitis, with or without blood tests that confirm it, it will be managed as acute pancreatitis. Acute means it's not chronic. It is serious and comes on quickly.

## Managing acute pancreatitis

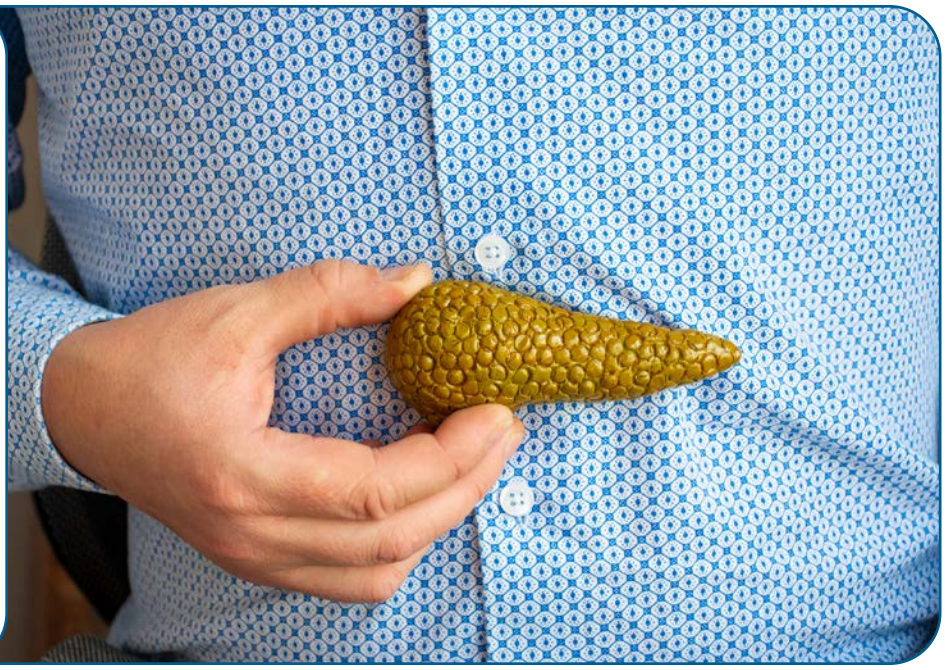
If you have acute pancreatitis, you'll be treated in the hospital. You'll be given fluids and medicine for pain relief.

Decisions about managing this serious issue will be led by a gastroenterologist and a doctor who has special training in pancreatic conditions.

Acute pancreatitis can lead to shock (when blood doesn't flow as well as it should), a life-threatening problem. Your team will stop immunotherapy.

### The pancreas

**The pancreas is a large gland behind the stomach. During immunotherapy, the immune system can attack the pancreas. This causes damage that affects hormone levels.**



## Blood sugar and diabetes

High blood sugar (glucose) can happen during ICI treatment.

When blood sugar is high, the pancreas makes insulin to lower it. If your blood sugar is high, you may have tests to find the cause.

### Diabetes (type 1 and type 2)

High blood sugar can be related to type 1 or type 2 diabetes.

**Type 1 diabetes** is when the pancreas doesn't make enough insulin. Type 1 diabetes can be a rare side effect of immunotherapy treatment. People with type 1 diabetes will have it for life.

**Type 2 diabetes** is when the body doesn't use insulin well.

Your care team will test your blood and urine to learn if you have type 1 or type 2 diabetes. You'll likely be asked to fast (not eat for a period of time) before blood sugar testing.

People with type 1 diabetes:

- Will manage it with daily insulin therapy and blood sugar checks
- Will be referred to an endocrinologist right away for diabetes management
- Can continue their immunotherapy
- May be treated in the hospital

If you have high blood sugar and tests confirm that you do not have type 1 diabetes, your team will:

- Continue to monitor your blood sugar levels
- Continue immunotherapy
- Treat you with medicine and discuss diet and lifestyle changes that might help manage high blood sugar
- Determine if high-dose steroids (if you are taking them) or type 2 diabetes is causing the high blood sugar

**If you have high blood sugar and fatigue (severe tiredness) or other symptoms of hormone issues, you will be tested for hypophysitis with adrenal insufficiency.**

## Diabetic ketoacidosis

Diabetic ketoacidosis (DKA) is a blood sugar crisis. This means blood sugar becomes very high. Fat is broken down too quickly, and toxic acids build up in the blood and urine. It can be life-threatening if insulin is not received.

DKA can happen to people with type 1 or type 2 diabetes. Possible symptoms of DKA include:

- Extreme thirst
- Urinating (peeing) a lot
- Belly pain
- Vomiting
- Weakness
- Confusion
- Dry skin
- Dry mouth

- Increased heart rate
- Breath that smells fruity

### Managing DKA

Your immunotherapy will be paused until your blood sugar is under control. Your team will:

- Refer you to an endocrinologist right away
- Manage this condition in the hospital
- Start you on insulin, glucose monitoring, and possibly early use of a continuous glucose monitor
- Order blood tests often

You may be treated with IV fluids with or without potassium.

**For people who develop hypophysitis or type 1 diabetes, NCCN recommends:**

1. Wearing a medical alert bracelet.
2. Having enough insulin when travelling.
3. Having a discussion with an endocrinologist or your oncologist before any scheduled procedures.

## Side effects of the pituitary and thyroid glands

The **pituitary gland** is in the brain. It makes hormones that control other glands and body functions.

The **thyroid gland** is in the neck. It makes hormones that help regulate heart rate, digestion, body temperature, and many other body functions. Your thyroid levels will be checked with blood tests before and during immunotherapy.

### Inflamed pituitary

If the pituitary gland becomes inflamed, it does not make enough hormones. This is a rare condition called hypophysitis, which can lead to adrenal insufficiency. That means the adrenal glands, which interact with the pituitary gland, don't make enough of the hormone cortisol.

Cortisol helps the body respond to stress, fight infections, and regulate blood sugar. A lack of cortisol can lead to permanent damage to the pituitary.

When the adrenal glands aren't making enough cortisol, symptoms can include:

- Fatigue
- Confusion
- Low blood pressure
- Loss of appetite
- Nausea
- Dizziness

Testing for pituitary issues includes blood tests and sometimes an MRI of the brain.

### Managing hypophysitis

For this condition, hydrocortisone is given to replace cortisol. This treatment may be needed for the rest of the person's life.

Immunotherapy will be paused until symptoms go away and hormone replacement is started.

When hormones have been replaced, ICI treatment can often be restarted.

Testosterone or estrogen may also be given. High-dose steroids may be prescribed in certain cases.

Normally, the adrenal glands make much more cortisol at certain times. This happens when there is infection, illness, or trauma to the body. To make up for the additional cortisol in these cases, hydrocortisone may be increased for a period of time.

### Thyroiditis and thyrotoxicosis

When the thyroid gland becomes inflamed, it's called thyroiditis. It's a painless condition, but sometimes it can lead to thyrotoxicosis, which can lead to hypothyroidism.

People with thyrotoxicosis have too much thyroid hormone but often have no symptoms. If they do, symptoms include heart palpitations, anxiety, weight loss, shaking, and sensitivity to heat.

### Managing thyrotoxicosis

If you have no symptoms of this condition, immunotherapy will be continued. If you do have symptoms, you'll be treated with beta-blockers that lower your heart rate and blood pressure until symptoms get better.

After 4 to 6 weeks, thyroid hormone levels may return to normal. But they may drop. Sometimes the level drops too low. This is called hypothyroidism.

### Hypothyroidism

Hypothyroidism, or underactive thyroid, doesn't always produce symptoms. If it does, the most common ones are:

- Weight gain
- Constipation
- Dry skin
- Sensitivity to cold

If there are symptoms, underactive thyroid may be treated with levothyroxine (Unithroid, Synthroid, Levoxyl, Levo-T), a medicine that replaces thyroid hormone. Your immunotherapy can be continued.

## Key points

- Treatment with immune checkpoint inhibitors (ICIs) can cause side effects of the endocrine system, where hormones are made.
- High levels of the enzyme lipase can be a sign of pancreatitis, an inflamed pancreas.
- If pancreatitis is acute or very serious, it will be treated in the hospital with IV hydration and pain relief, and immunotherapy will be stopped.
- High blood sugar can be related to diabetes.
- Although rare, diabetes can develop because of ICI treatment and may need to be managed for life.
- If the pituitary gland becomes inflamed and the adrenal glands don't make enough cortisol, hydrocortisone is used to replace it.
- If the thyroid is not working well, it's treated with levothyroxine, a thyroid hormone replacement.

## Questions to ask

- Will I need to continue seeing an endocrinologist after immune checkpoint inhibitor treatment?
- Why am I having imaging tests?
- What do I need to do to manage diabetes?
- How do I get a medical alert bracelet?
- Does hormone replacement put me at risk for other issues?

## What's next?

In Chapter 8, side effects of the lungs and heart will be explained. You can learn about what these side effects are and how they are managed.

# 8

# Side effects of the lungs and heart


59 Inflamed lungs

61 Inflamed heart and vessels

63 Key points

63 Questions to ask

In this chapter, you'll learn about side effects of immune checkpoint inhibitors (ICIs) that affect the lungs and heart.

You'll read about how these conditions are confirmed and how they are managed. You'll also learn if and how they will affect your immunotherapy treatment.

## Inflamed lungs

Although uncommon, treatment with immune checkpoint inhibitors can cause the lungs to become inflamed. This is called pneumonitis. Symptoms include:

- Trouble breathing
- Dry cough
- Fever
- Chest pain

Pneumonitis can usually be seen on a CT scan, so you'll likely have one.

### Managing inflamed lungs

The first sign of pneumonitis is often a low level of oxygen in the blood. If pneumonitis is:

- **Mild**, it doesn't cause symptoms and only affects 1 lung lobe or a small part of lung tissue. Your team may pause immunotherapy.
- **Moderate**, it means you have symptoms or your symptoms are getting worse
- **Severe**, your symptoms:
  - Involve all lobes and more than half of lung tissue
  - Make it hard to care for yourself
  - Affect your ability to breathe
- **Life threatening**, it may lead to lung failure or death

**Checking your blood oxygen level**

**A pulse oximeter measures the amount of oxygen being carried by red blood cells. It does this by shining light through your finger.**

## Managing mild pneumonitis

In addition to a physical exam, you'll have an oxygen level test. The level will be checked when you are at rest and after movement.

You may have a chest CT with contrast to see if an infection is causing the symptoms, and not ICI treatment. Contrast is used to make the pictures of the inside of the body clearer. Another chest CT may be needed in 4 to 6 weeks.

## Managing moderate pneumonitis

If pneumonitis symptoms are moderate, your team will pause immunotherapy.

Your care team:

- Will prescribe an oral or IV steroid.
- May discuss management with a pulmonologist.
- May prescribe oral antibiotics (if infection is likely).
- May swab your nose to test for viruses.
- May test your mucus, blood, and urine for bacterial and fungal infections.
- Will order a chest CT with contrast (and another in 3 to 4 weeks).
- May order a bronchoscopy with bronchoalveolar lavage and biopsy. These tests can help check for lung infection.
- May start mycophenolate mofetil.
- Will do a follow-up physical exam and oxygen level test every 3 to 7 days.

If symptoms don’t get better in 2 to 3 days after steroid treatment, pneumonitis will be treated as severe.

## Managing severe or life-threatening pneumonitis

If pneumonitis symptoms are severe, your team will pause immunotherapy.

You care team:

### Bronchoalveolar lavage

**During a bronchoscopy, a small amount of sterile saline (saltwater) may be put into the lungs. The saline washes the airway and is then suctioned back through the tube. The fluid is tested for infection.**

- Will prescribe an oral or IV steroid.
- May prescribe oral antibiotics (if infection is likely).
- Will treat you in the hospital.
- Will discuss your management with experts in pulmonology and infectious disease.
- Will order a bronchoscopy with bronchoalveolar lavage and biopsy. These tests can help check for lung infection.
- May swab your nose to test for viruses.
- May test your mucus, blood, and urine for bacterial and fungal infections.
- May have you see a cardiologist to learn if symptoms are related to an issue with your heart.

If your symptoms don't get better after 2 days, your team may start:

- IVIG
- Tocilizumab
- Mycophenolate mofetil
- Infliximab

The first 2 treatments are preferred.

### Other specialists you may see

These doctors may also be involved in your care:

- A pulmonologist, an expert in lung diseases
- An infectious disease doctor, an expert in bacterial, viral, and fungal infections
- A cardiologist, an expert in diagnosing and treating heart conditions

## Inflamed heart and vessels

ICI treatment may affect the heart and blood vessels.

The heart may become inflamed. This is called myocarditis. The heart becomes bigger (swollen) and weaker. Such side effects are rare but can come on quickly and become very severe. Severe cases can lead to changes to the heartbeat and heart failure.

Myocarditis symptoms include:

- Chest pain
- Irregular heartbeat
- Shortness of breath
- Fatigue
- Fainting
- Weakness

Symptoms may happen as early as days to weeks after 1 to 2 doses of ICI.

If you have symptoms, you'll have a series of tests to look for heart and blood vessel issues. The goal of testing is to find these issues early and learn if they are due to immunotherapy or some other heart or blood vessel disease.

You'll see a cardiologist or cardio-oncologist right away. A cardiologist is an expert in heart conditions. A cardio-oncologist helps people with cancer manage heart conditions.

How myocarditis is managed will depend on how severe it is and if symptoms respond to steroid treatment.

You may be admitted to the hospital for closer monitoring, more testing, or treatment.

## Managing inflamed heart

If myocarditis is confirmed, your immunotherapy will be stopped. You'll be given high-dose IV steroids for 3 to 5 days. The goal of steroid therapy is to reverse the immune attack against the heart, support and recover heart function, and prevent myocarditis from coming back.

If your symptoms are getting better, you'll be switched to an oral steroid. This will be tapered over 6 to 12 weeks based on how symptoms respond and if certain features improve.

## If symptoms don't respond to steroids

If myocarditis doesn't improve within 1 to 2 days of starting steroid treatment, you'll be cared for and monitored in an intensive care unit, or ICU. You'll begin treatment with immunosuppressants. Therapies are listed in alphabetical order in **Guide 19**. Other heart and blood vessel issues include:

- Pericarditis, when the outer lining of the heart (pericardium) becomes inflamed
- Pericardial effusion, when fluid builds up around the heart
- Vasculitis, when blood vessels (arteries and veins) become inflamed

Your care team may do tests to see if myocarditis is part of the cause of these issues. Your team will treat your symptoms according to your test results.

**Guide 19**
**Myocarditis treatment if steroids don't help**

| |
|---|
| Abatacept (Orencia)* |
| Alemtuzumab |
| Antithymocyte globulin (Atgam) |
| Infliximab** |
| IVIG |
| Methotrexate |
| Mycophenolate mofetil |
| Plasma exchange |

* Has been used with ruxolitinib (Jakafi) when myositis (inflamed muscles) is diagnosed with myocarditis.

** Used with caution in people with certain heart conditions.

## Key points

- Pneumonitis, or inflamed lungs, can cause trouble with breathing, a dry cough, and fever.
- Inflamed lungs are treated with steroids and possibly antibiotics.
- Pneumonitis can be severe and require treatment in the hospital.
- Myocarditis is an inflamed heart. The outer lining of the heart can also become inflamed. This is called pericarditis.
- Symptoms of an inflamed heart include chest pain, extreme tiredness, and fainting.
- Vasculitis is when the arteries and veins become inflamed.
- Myocarditis is treated with high-dose steroids. Immunosuppressants may be needed if myocarditis persists.

## Questions to ask

- What should I do if I miss a steroid dose?
- How do I prepare for a bronchoscopy?
- Will I need to continue to see a pulmonologist after ICI treatment?
- Heart disease runs in my family. Am I at higher risk for a heart attack during treatment or after?
- Should I see a cardio-oncologist?

## What's next?

The next chapter explains side effects of ICI treatment of the eyes. One common side effect is dry eyes. This can be managed with over-the-counter eye drops or prescription eye medicine. Other vision- and eye-related side effects and management are also discussed.

# 9
# Side effects of the eyes

65 Managing eye-related side effects

68 Key points

69 Questions to ask

Treatment with immune checkpoint inhibitors (ICIs) can cause side effects of the eyes and affect vision. Dry eyes are a common eye-related side effect of ICI therapy. Over-the-counter eye drops (artificial tears) can help with dryness and soothe irritation.

In this chapter, you'll read about eye-related symptoms and how eye conditions are managed (see **Guide 20**). You'll also find out if your symptoms will affect your immunotherapy treatment.

## Managing eye-related side effects

If your vision changes or you have other eye-related symptoms, your care team will refer you to an ophthalmologist for a complete eye exam. Ophthalmologists are doctors who specialize in vision and eye diseases. The ophthalmologist will likely guide the care and treatment of your eyes.

The ophthalmologist will use a special microscope with a very bright light (called a slit lamp) to examine your eyes. The eye exam tests:

- Vision and color vision
- Pupil health
- Eye pressure
- The front of the eye
- The retina and retinal nerve fibers
- The macula

Many types of infections and diseases can cause eye problems. Compared to these, eye conditions due to immunotherapy are very uncommon. So you'll have blood tests to rule out these other possible causes of your symptoms.

Your health care team may recommend that you avoid wearing contact lenses—if you normally wear them. It's a good idea to avoid using eye makeup, too.

### Uveitis

Uveitis is a common side effect of ICI treatment. The uvea is the layer of tissue beneath the white of the eye. When the uvea becomes inflamed, it's called uveitis. The eye usually becomes red.

| Guide 20<br>Eye-related symptoms |
|---|
| Blurred, distorted, or double vision |
| Blind spots |
| Change in color vision |
| Sensitivity to light |
| Tenderness or pain |
| Eyelid swelling |
| Bulging eyes |
| Itchy eyes |
| Redness |

Uveitis can appear at the front of the eye (anterior uveitis), at the back of the eye (posterior uveitis), or sometimes all over the eye (panuveitis) (see **Guide 21**).

Anterior uveitis is a more common side effect of immunotherapy than posterior uveitis. Posterior uveitis is more serious.

Steroids are the mainstay of uveitis treatment. If a strong steroid doesn't improve symptoms of **panuveitis,** your ophthalmologist may add:

- Infliximab
- An antimetabolite, such as methotrexate

## Episcleritis

The episclera is the thin layer of clear tissue that covers the white of the eye (the sclera). If the episclera becomes inflamed, it doesn't usually affect eyesight. Red or purple discoloration of the eye is usually a symptom.

Treatment is not always needed. Artificial tears can help with symptoms. If there are changes in eyesight, immunotherapy may be paused. Treatment may include non-steroidal anti-inflammatory eye drops, steroid eye drops, or an oral or IV steroid.

## Scleritis

The sclera is the white of the eye. It protects the eye and helps maintain its structure. When the sclera is inflamed, it's often painful. Eyes may be tender, red, and swollen.

Mild scleritis does not affect vision. It is treated with the oral non-steroidal anti-inflammatory drugs (NSAIDs) flurbiprofen or indomethacin (Indocin). Oral or IV steroids are needed for scleritis that is causing vision changes. Immunotherapy will be paused until the scleritis goes away.

**Guide 21**
**Managing uveitis**

| | |
|---|---|
| **Front of the eye (anterior)** | • If mild, immunotherapy will continue with monitoring.<br>• If moderate, immunotherapy will be paused until the eye is stable.<br>• Steroid eye drops, an oral or IV steroid, or both will also be prescribed. |
| **Back of the eye (posterior)**<br>**or**<br>**All of the eye (panuveitis)** | • Immunotherapy will be paused.<br>• Steroid eye drops and an oral or IV steroid will be prescribed. |

## Keratitis

The clear tissue on the front of the eye, called the cornea, can also become inflamed. This is called keratitis. Treatment is not needed if the inflammation is mild, and immunotherapy can continue. Artificial tears can help with symptoms.

Moderate or severe keratitis can be painful and may affect eyesight. See **Guide 22** to learn how it's managed.

If changes in eyesight occur, symptoms can be treated with topical steroid eye drops (prednisolone) or immunosuppressant eye drops (cyclosporine).

If vision gets worse or a hole emerges in the cornea, your ophthalmologist may add:

- Serum tears that are made from a person's own blood
- A protective contact lens that shields the cornea while it heals

## Maculopathy

Maculopathy is damage to the center part of the retina, called the macula. This condition causes blurry central vision. Central vision is what we use to read, drive, and recognize faces. Treatment will include steroid eye drops (prednisolone) or an oral or IV steroid. Your immunotherapy will be paused.

**Guide 22**
**Managing keratitis**

| | |
|---|---|
| **Moderate symptoms** | • Steroid eye drops<br>• Immunosuppressant eye drops |
| **Severe symptoms** | • Steroid eye drops<br>• Immunosuppressant eye drops<br>• Serum tears<br>• Protective contact lens |

### Complete eye exam

**If you have symptoms or vision changes, your care team will refer you to an ophthalmologist for an eye and vision exam. A special microscope with a bright light (a slit lamp) is a key part of the test.**

### Blepharitis

Blepharitis is when eyelids are inflamed. They may be red and swollen. Your eyes may tear and burn. If symptoms are mild or only slightly affect your vision, artificial tears, warm compresses on the eyelids, or both will be recommended. Immunotherapy can continue.

If symptoms are more severe, treatment will include an oral antibiotic (doxycycline) or an antibiotic ointment. Immunotherapy will be paused.

### Orbital myositis

The orbital muscles control eye movement. They are in the eye's orbit, or socket. Orbital myositis is when the muscles in the orbit become inflamed. Other issues that affect these muscles include thyroid-like eye disease and cranial neuropathy.

Immunotherapy is paused. These conditions are treated with an oral or IV steroid until symptoms get better.

### Optic neuritis

This is when the optic nerves become inflamed. These nerves send signals from the retina to the brain. Immunotherapy will be paused. Treatment will include oral or IV steroids.

## Key points

- An ophthalmologist will examine your eyes if you have symptoms that affect your vision or eye health. The ophthalmologist may guide your care and further decisions about immunotherapy.
- Eye-related symptoms caused by immunotherapy include blurry vision, pain, eye redness or swelling, and itching.
- Dry eyes are a common eye-related side effect of immune checkpoint inhibitor (ICI) ICI therapy.
- If dry eye is mild, it can be treated with over-the-counter eye drops (artificial tears).
- Serious eye-related side effects are often treated with steroid eye drops or with an oral or IV steroid.
- Uveitis is inflammation of the uvea—the tissue beneath the white of the eye. It's treated with steroids.
- If eye-related symptoms are serious, immunotherapy will be paused.

## Questions to ask

- How often can I use artificial tears for dry eye symptoms?
- How long can I use steroids for eye inflammation and pain?
- What should I do if I miss a dose of the steroid?
- Do over-the-counter and prescription eye drops cause any side effects?
- Can eye-related side effects spread to other parts of my body?

## What's next?

In the next chapter, you'll read about rare but serious side effects of the brain, spinal cord, and nerves. Learn about symptoms, tests, and procedures you may have, and how symptoms are managed.

# 10
# Side effects of the nervous system


71 Tests of the nervous system

71 Managing nervous system side effects

71 Myasthenia gravis

72 Guillain-Barré syndrome

73 Meningitis (non-infectious)

73 Brain inflammation

74 Disease of the myelin sheath

74 Nerve damage

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Treatment with immune checkpoint inhibitors (ICIs) can cause nervous system side effects. The nervous system includes the brain, spinal cord, and nerves throughout the body. Side effects of the nervous system are rare, but they can be very serious.

This chapter describes immunotherapy-related conditions of the nervous system, tests you may have, and how these side effects will be managed.

## Tests of the nervous system

You'll have blood tests and possibly an MRI scan of the brain, spine, or both. Contrast materials may be used. Contrast helps enhance images of the organs and structures in the body. You may also have tests for:

- Heart function (electrocardiogram)
- Lung function and breathing tests
- Nerve function tests (electromyography and nerve conduction study)
- Disease of the spinal fluid and spinal tissues (lumbar puncture)

## Managing nervous system side effects

Your care and treatment will likely include a neurologist, a doctor who is an expert in conditions of the brain and nervous system.

Managing side effects will often include an oral or IV steroid. It may also include:

- Plasmapheresis, when plasma gets removed from the blood and replaced
- IVIG
- Medicine to help with pain

Each condition of the nervous system and its testing and treatment for side effects is explained in the rest of this chapter

## Myasthenia gravis

Symptoms of myasthenia gravis include:

- Muscle weakness
- Double vision
- Drooping eyelids
- Difficulty speaking
- Weak facial muscles
- Weak breathing muscles

If you have myasthenia gravis, immunotherapy will be stopped. You may be admitted to the hospital even if your symptoms are mild. This is because myasthenia gravis can worsen quickly.

Moderate symptoms make it hard to care for yourself. You'll be given a low-dose oral steroid to start. The dose will be adjusted as needed. You'll also be given pyridostigmine (Mestinon), a muscle-strengthening medicine.

If you have myasthenia gravis, your team will also test to see if you have myocarditis, myositis, or both. These conditions can

happen at the same time. For more on myocarditis, see *Chapter 8: Side effects of the lungs and heart*. For myositis, see *Chapter 11: Side effects of the muscles and joints*.

### Managing severe myasthenia gravis

Symptoms are severe if they:

- Limit your ability to care for yourself
- Make walking hard
- Make it hard to swallow
- Make your face weak
- Get worse quickly

If symptoms are severe, you will:

- Be admitted to the hospital (possibly intensive care)
- Be treated with IV steroid
- Have plasmapheresis or IVIG
- Have your lungs checked often
- Be seen by a neurologist every day
- Have your medicines adjusted as needed

You'll need to be admitted to the hospital, where you'll be treated with an IV steroid and plasmapheresis or IVIG. If these treatments don't help, you may be treated with rituximab.

Certain medicines can worsen myasthenia gravis. If you're taking any of them, your care team will ask you to stop. These include beta-blockers, certain antibiotics, and IV magnesium.

## Guillain-Barré syndrome

Guillain-Barré syndrome is a nerve condition that causes weakness of the arms, legs, face, breathing muscles, and eye nerves.

The first symptom is often pain in the lower back and thighs. This syndrome is treated in the hospital. Immunotherapy will be stopped.

Management of Guillain-Barré syndrome is outlined in **Guide 23**.

You'll be treated with an IV steroid and plasmapheresis or IVIG. Gabapentin, pregabalin, and duloxetine are used to relieve pain.

Other nervous system issues can happen at the same time as Guillain-Barré syndrome. Your care team will monitor you for symptoms of other conditions.

**Guide 23**
**Managing Guillain-Barré syndrome**

| | |
|---|---|
| **You will:** | • Be admitted to the hospital (possibly intensive care)<br>• Be treated with an IV steroid<br>• Have plasmapheresis or IVIG<br>• Have your lungs checked often<br>• Be seen by a neurologist often<br>• Be given 1 of the following:<br>  • Gabapentin<br>  • Pregabalin<br>  • Duloxetine (Cymbalta) |

## Meningitis (non-infectious)

The meninges are the thick layers of tissue that cover the brain and spinal cord. Meningitis happens when the meninges become inflamed. Non-infectious means that the inflammation is not caused by bacterial infection. If bacteria are found, management of meningitis will be adjusted.

Symptoms of meningitis include:

- Headache
- Sensitivity to light
- Neck stiffness
- Possible fever, vomiting, or both

If you have meningitis, immunotherapy will be paused or stopped. If your symptoms are making it hard to care for yourself, you'll be admitted to the hospital.

You may be given IV antiviral medicine while your care team waits for test results to make sure you don't have a bacterial or viral infection. You'll be treated with an oral steroid.

For more severe symptoms, you'll be given an IV steroid, which will be tapered over 4 weeks once symptoms have gone away.

## Brain inflammation

Brain inflammation, or encephalitis, is an uncommon but possible side effect of immune checkpoint inhibitors. Symptoms range from mild confusion to serious brain function problems. Symptoms include:

- Confusion
- Changes in behavior
- Headaches
- Seizures
- Short-term memory loss
- Problems reacting or responding
- Trouble speaking

### Managing inflamed brain

You'll be treated in the hospital if your symptoms are making it hard to care for yourself. Your immunotherapy will be paused or stopped.

You may be given IV antiviral medicine while your care team waits for test results to make sure you don't have a bacterial or viral infection. You may be seen by infectious disease experts.

Your team will start an IV steroid and taper the dose as your symptoms get better.

If symptoms become severe or get worse over 24 hours, you may be treated with a higher dose of IV steroid and IVIG or plasmapheresis.

If symptoms don't improve in 1 to 2 weeks or a certain protein is found in the blood, treatment with rituximab may be started.

## Disease of the myelin sheath

The myelin sheath is the protective covering of the nerve fibers. Disease of the myelin sheath is any condition that causes damage to this covering. These conditions can affect the nerves in the brain and spinal cord, and the nerves that lead to the eyes. Conditions include:

- **Transverse myelitis**, which can cause pain, muscle weakness in the legs and arms, sensory and reflex problems, and bladder or bowel changes.
- **Acute demyelinating encephalomyelitis**, which can cause the same side effects as brain inflammation. Additional symptoms include weakness of a specific area of the body, numbness or tingling, loss of balance, lack of coordination, and vision loss.
- **Optic neuritis**, or inflamed optic nerves, which can cause vision loss and changes, eye pain, and nerve swelling. If you have optic neuritis, an ophthalmologist will oversee your treatment. An ophthalmologist is a doctor who specializes in vision and eye diseases.

If you have any of these 3 conditions:

- You'll be treated in the hospital
- Your immunotherapy will be stopped
- You'll be given treatment with an IV steroid for 3 to 5 days

If symptoms don't improve or get worse after 2 days of IV steroid treatment, IVIG or plasmapheresis may be started.

## Nerve damage

Nerve damage is also called neuropathy. Your care team will evaluate your symptoms to make sure that something other than ICI treatment isn't the cause.

Neuropathy can affect 2 or more nerves. Rare, serious cases can affect the stomach muscles so that the stomach doesn't empty the right way.

Symptoms are treated as:

- **Mild**, if they don't make it hard for you to manage your daily routine and don't bother you. Your care team may pause immunotherapy.
- **Moderate**, if they make it hard for you to care for yourself and they bother you. You may have pain but not weakness or trouble walking. Immunotherapy will be paused.
- **Severe**, if you have a lot of trouble caring for yourself, as well as weakness and loss of feeling that makes it hard to walk.

### Managing nerve damage symptoms

Mild symptoms are monitored for a week to see if pain, weakness, trouble walking, or other new symptoms are affecting your day-to-day life.

If there are any issues of the cranial nerves, symptoms are treated as moderate, even if they are mild. Cranial nerves stem directly from the brain. They help us see, hear, taste, smell, and feel sensations.

If symptoms are moderate or go from mild to moderate after a week of monitoring, you'll be treated with an oral steroid. If symptoms continue to get worse, an IV steroid will be started. Pain will be treated with gabapentin, pregabalin, or duloxetine.

Severe symptoms are managed like those of Guillain-Barré syndrome (see **Guide 23**).

## Key points

- Side effects of the brain, spinal cord, and nerves—the nervous system—are rare, but they can be serious.
- Tests for nervous symptom side effects will include imaging and may include heart and lung function tests and a lumbar puncture.
- A neurologist, a doctor who is an expert in conditions of the brain, spinal cord, and nerves, will likely be involved in your care and treatment decisions.
- Symptoms of nervous system problems can include headaches, weakness, difficulty speaking, and issues with vision.
- Nervous system symptoms are often treated with IVIG, plasmapheresis, and medicine for pain relief.
- People with severe nervous system side effects are often treated in the hospital.
- Your immunotherapy will likely be paused or stopped so your nervous system condition can be treated.

## Questions to ask

- If I have a headache, should I contact my care team?
- How will I know if I should go to the ER?
- Will my symptoms go away after immunotherapy ends?
- Will I need to see a neurologist after immunotherapy ends?
- Do I have an increased risk for stroke during ICI treatment?

## What's next?

Side effects of the muscles and joints, including inflammatory arthritis, that can happen during ICI therapy are explained in the next chapter. Management of these side effects will also be discussed.

# 11
# Side effects of muscles and joints


77 Inflammatory arthritis

78 Polymyalgia rheumatica

79 Giant cell arteritis

80 Muscle weakness

81 Key points

81 Questions to ask

Treatment with immune checkpoint inhibitors (ICIs) can cause joints, tendons, ligaments, bones, and muscles to become inflamed. In this chapter, you'll read about these side effects, their symptoms, and how they are managed.

If you have any side effects of your muscles or joints, a rheumatologist is likely to be involved in your care. A rheumatologist is a doctor who's an expert in managing arthritis and other muscle and joint issues.

**Guide 24** lists some of the medicines your health care team may prescribe if you have a side effect in your muscles and joints.

## Inflammatory arthritis

**Inflammatory arthritis** is when joints become inflamed and damaged. A reaction to immune checkpoint inhibitors (ICIs) is 1 of the many causes of inflammatory arthritis.

It can affect 1 or 2 joints or many joints. Symptoms include:

- Joint swelling
- Joint pain
- Stiffness

Symptoms may also be like those of polymyalgia rheumatica (discussed in the next section).

Your care team will examine the swollen joints. You may have imaging tests of the affected joints.

How symptoms are managed depends on how many joints are affected, how severe the swelling is, and if you're having a hard time caring for yourself because of your symptoms. A rheumatologist may inject a steroid into the affected joints with a needle.

### Managing mild inflammatory arthritis

Symptoms of inflammatory arthritis are considered mild if they only affect 1 or 2 joints. Pain can usually be managed with non-steroidal anti-inflammatory drugs (NSAIDs)

**Guide 24**
**Medicines that treat muscle and joint symptoms**

| | |
|---|---|
| **DMARD (disease-modifying anti-rheumatic drug)*** | • Hydroxychloroquine<br>• Leflunomide (Arava)<br>• Methotrexate<br>• Sulfasalazine (Azulfidine) |
| **IL-6 inhibitor** | • Sarilumab<br>• Tocilizumab |
| **TNF inhibitor** | • Adalimumab<br>• Certolizumab pegol<br>• Etanercept<br>• Golimumab<br>• Infliximab |

* You will have blood tests first to make sure a DMARD is safe for you.

like ibuprofen (Advil), aspirin, and naproxen (Aleve).

Your care team may:

- Use a needle to take fluid from a joint to be tested for a viral infection that may be causing the symptoms.
- Prescribe a COX2 inhibitor (a kind of NSAID) or gabapentin and pregabalin for pain relief.
- Prescribe oral steroids if NSAIDs don't help. Or prescribe a disease-modifying anti-rheumatic drug (DMARD) if steroids can't be tapered. DMARDs are used to treat autoimmune diseases.
- Pause immunotherapy if treatment with NSAIDs, steroids, and DMARDs don't help.

### Managing moderate or severe inflammatory arthritis

If symptoms are **moderate**, at least 1 joint is severely inflamed. Your immunotherapy may be paused.

If symptoms are **severe**, many joints are affected and you're having a hard time caring for yourself. Your immunotherapy will be paused. It may be stopped if symptoms are making it hard to do your daily routine.

Your care team will prescribe an oral steroid to be taken until symptoms improve. The steroid will then be tapered over 4 to 6 weeks. If the steroid can't be tapered in a week, your team may prescribe 1 of the following:

- A DMARD
- An IL-6 inhibitor
- A TNF inhibitor

If symptoms suggest another type of arthritis, apremilast or another biologic medicine may be prescribed.

## Polymyalgia rheumatica

Polymyalgia rheumatica (PMR) is a disease that causes muscle aches, pain, and stiffness. The shoulders and hips are most often affected, but pain and stiffness can happen in other areas, too. Symptoms usually occur in the morning. They can make it hard to do your regular daily routine or care for yourself.

PMR can happen with other types of arthritis and is related to giant cell arteritis (GCA), another inflammatory condition. GCA is explained in the next section. Here are some things to know about PMR and GCA:

- Blood tests will be ordered to check levels of inflammation in your body.
- Both conditions can affect vision.
- If you have double or blurry vision, or vision loss, PMR is treated similarly to GCA.
- You'll be treated with a steroid.
  - If steroid (more than 20 mg) treatment continues for more than 4 weeks, you are at risk for a certain kind of pneumonia that is caused by fungal infection. You'll be given treatment to help prevent this pneumonia.
- Management may include an IL-6 inhibitor.
  - If you have a history of colon disease, you may be at increased risk for a colon tear if you take an interleukin (IL)-6

inhibitor. Your team will use caution if using an IL-6 inhibitor.

### Managing PMR

Your immunotherapy may continue. If your vision is affected, immunotherapy will be paused until you can be evaluated for GCA.

Your team will prescribe an oral steroid that's tapered over 6 to 8 weeks. If your symptoms aren't improving, your team may:

- Increase the steroid dose
- Pause immunotherapy
- Treat with a DMARD
- Treat with an IL-6 inhibitor if steroids cannot be tapered or your symptoms aren't getting better

## Giant cell arteritis

GCA can affect vision. Blood vessels on the sides of the forehead (temporal arteries) narrow and become inflamed. If untreated, this condition can lead to blindness.

In addition to vision loss or double or blurry vision, symptoms include:

- Scalp tenderness
- Jaw pain when chewing
- Fevers, night sweats, and weight loss

If you have symptoms, you'll be treated with an oral steroid. Your care team will have you see an ophthalmologist (a doctor who specializes in vision and eye disease) or a vascular surgeon (an expert in conditions of veins and arteries) right away. You'll have a temporal artery biopsy, ultrasound, or both.

Your team:

- Will pause immunotherapy
- Will start or continue oral steroid treatment, which will be tapered over 2 to 3 months

**Polymyalgia rheumatica (PMR)**

**PMR causes muscle aches, pain, and stiffness, especially in the shoulders.**

- May treat with IL-6 inhibitor
- May treat with an IV steroid for 3 days, followed by an oral steroid, which will tapered

## Muscle weakness

ICI treatment can cause muscle weakness, called myositis. It most often affects muscles in the neck, upper arms, shoulders, hips, and thighs. You may also have muscle pain (myalgia), trouble swallowing, or both.

Muscle strength testing can help find nervous system problems that may be causing the weakness. Your team will order an electrocardiogram (which may show damage to the heart) and blood tests to check muscle enzyme levels (which may show muscle damage).

You may also have:

- Electromyography (measures muscle and nerve function)
- MRI of the affected muscles
- A muscle biopsy

### Managing mild or moderate symptoms

If symptoms are mild or moderate, you may find it hard to care for yourself because of pain, weakness, or both. Blood testing may show high levels of muscle enzymes.

Your team may pause or stop your immunotherapy. Treatment will include an oral steroid, and muscle enzyme levels may be monitored.

**Myositis can happen at the same time as:**

**Myocarditis**
(*Chapter 8: Side effects of the lungs and heart*)

**Myasthenia gravis**
(*Chapter 10: Side effects of the nervous system*)

If you have myositis, your team will test to see if you also have myocarditis, myasthenia gravis, or both. Treatment will be based on those results.

If steroid treatment doesn't help, your team may retest you for myasthenia gravis and myocarditis. Your symptoms will be treated as severe or life-threatening.

### Managing severe or life-threatening symptoms

Immunotherapy will be paused or stopped, and you'll be treated in the hospital.

You may be given an IV steroid for 3 days followed by an oral steroid for 4 weeks. The oral steroid will be tapered.

If steroid treatment doesn't help after 2 to 4 weeks, you'll also be given a DMARD.

If you have trouble swallowing, other life-threatening issues, or symptoms that don't improve after a week, you may receive 1 of the following:

- IVIG
- Methotrexate or rituximab
- Mycophenolate mofetil

Mycophenolate mofetil or azathioprine may be substituted for methotrexate in certain cases. Muscle enzyme levels will be monitored until symptoms have gone away and steroids have been stopped.

If myositis happens with myocarditis and myasthenia gravis at the same time, you'll be treated with abatacept with ruxolitinib.

## Key points

- Immune checkpoint inhibitor (ICI) therapy can cause joint pain and stiffness.
- Mild inflammatory arthritis can be treated with over-the-counter pain relievers, but more serious symptoms require steroid treatment.
- Polymyalgia rheumatica (PMR) and giant cell arteritis can affect a person's vision.
- Treatment for PMR and giant cell arteritis (GCA) includes steroids and sometimes other medicines like interleukin (IL)-6 inhibitors.
- Myositis causes muscle weakness, which can lead to trouble swallowing.
- Myositis can be very severe and require in-hospital treatment.
- Treatment for myositis may include IV steroids and other IV medicines, especially if swallowing is difficult for you.

## Questions to ask

- Will muscle and joint issues go away when I finish immunotherapy treatment?
- Why am I taking medicines in addition to steroids?
- What should I do if I miss a dose of steroids or other therapies?
- Do DMARDs cause side effects?

## What's next?

ICI can cause issues with the blood and bone marrow. Your body may have conditions that relate to red blood cells, white blood cells, blood platelets, or all 3. How these side effects are managed will be discussed in the next chapter.

# 12
# Side effects of blood and bone marrow

83 Anemia

85 Thrombocytopenia (low blood platelets)

86 Hyperinflammation

87 Key points

87 Questions to ask

Treatment with immune checkpoint inhibitors (ICIs) can lead to side effects of the blood and bone marrow.

In this chapter, you'll learn about conditions of the blood and bone marrow and how they are managed. If you have any of these side effects, a hematologist, a doctor who's an expert in blood and bone marrow disorders, will likely be involved in your care.

## Anemia

Anemia happens when the body doesn't have enough red blood cells. Red blood cells contain hemoglobin, which carries oxygen to the body's tissues. Red blood cells are made in the bone marrow (the soft tissue in the bones).

If your red blood cell count is low, your care team will:

- Order blood tests to learn why
- Check levels of certain vitamins and minerals
- Test to see if there is blood in your stool (poop)
- Watch to see how long red blood cell levels are low
- Order a lab test of your bone marrow tissue (biopsy)(see figure on page 85)

Causes other than immune checkpoint inhibitors, including viruses, bacteria, or other things you may have had contact with, will be ruled out. The results of these tests will determine how anemia is managed (discussed later in this chapter).

**If hemolytic anemia becomes life-threatening, inpatient care is necessary.**

If anemia is confirmed, your immunotherapy will be paused or stopped.

There are many types of anemia. The 2 types that can ICI therapy are:

- **Aplastic anemia**, when the body doesn't make enough red blood cells.

  This can occur with or without low counts of white blood cells and platelet-s. Low platelet counts will be discussed later in this chapter.

- **Hemolytic anemia**, when the red blood cells are being destroyed faster than the bone marrow is making them.

### Managing aplastic anemia

If aplastic anemia is severe, counts of certain blood cells, platelets, or both are very low. You may be admitted to the hospital as an inpatient.

You may have blood transfusions to restore your red blood cells.

Growth factor support may be used. This means you'll be given medicine to help your bone marrow make blood cells.

You'll be given an oral or IV steroid until red blood cell counts improve. Steroids will be tapered over 1 or 2 months. If anemia does not improve after 7 days of steroid treatment, you may be treated with 1 of the following immunosuppressants:

- IVIG
- Cyclosporine
- Antithymocyte globulin
- Mycophenolate mofetil
- Tacrolimus

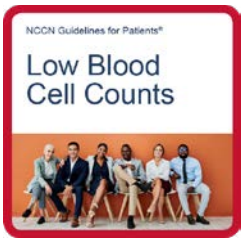
If these treatments don't help, your health care team may recommend a hematopoietic cell (stem cell) transplant (HCT) to replace damaged bone marrow cells with healthy cells.

## Managing hemolytic anemia

If red blood cell counts aren't dangerously low, anemia can be treated with an oral steroid instead of a blood transfusion until stable. The steroid will be tapered over 1 to 2 months.

If red blood cells become so low that a blood transfusion is needed, you may be admitted to the hospital. Treatment will also include steroids until your red blood cell counts improve. Steroids will be tapered over 1 or 2 months.

More information on anemia is available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.



### Transfusion

A transfusion is a common procedure to replace blood or blood components (red blood cells or platelets). It is given through an intravenous (IV) line, a tiny tube inserted into a vein with a small needle.

Rituximab may be used if steroids don't help after 5 to 7 days. If red blood cell counts become low enough to be life-threatening, rituximab may be added after 3 to 5 days of no improvement.

If steroids and rituximab don't help, you may be treated with 1 of the following:

- IVIG
- Tacrolimus
- Cyclophosphamide (chemoimmunotherapy)
- Mycophenolate mofetil
- Cyclosporine (only if life-threatening)

## Thrombocytopenia (low blood platelets)

Thrombocytopenia means having a low number of blood platelets. Platelets are cells that help the blood clot. When blood platelets become low as a side effect of ICIs, the condition is usually mild.

### Tests for thrombocytopenia

Your blood may be tested often. Tests will assess the health of your blood cells and how quickly your blood clots. Your team:

- Will test for certain infections and other conditions that may be causing low blood platelets
- Will order ultrasound imaging of your belly
- May order a bone marrow biopsy and aspiration. Bone marrow will be removed and sent to a lab for testing.

### Managing low blood platelets

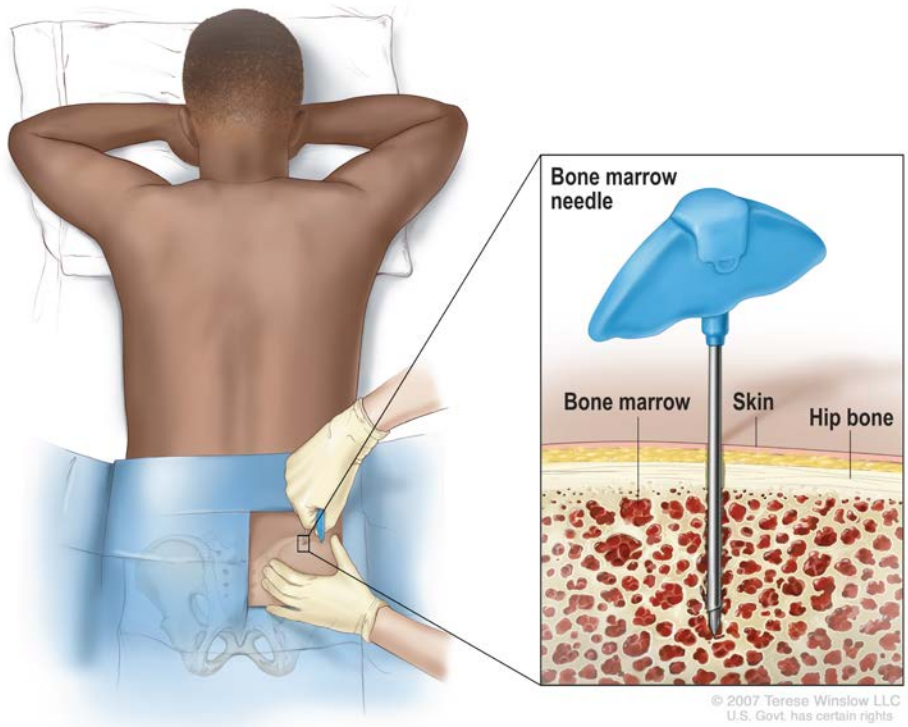
Low blood platelet counts may get better without treatment. If your condition is mild, your care team will monitor for signs of bleeding. Your immunotherapy will be continued. If there is bleeding because of low platelets, you may be treated with IVIG.

If your blood platelet count becomes very low, immunotherapy will be paused.

If your platelet count does not improve after 4 to 6 weeks of monitoring, you may be treated with oral or IV steroids for 2 to 4 weeks, then tapered over 4 to 6 weeks.

**Bone marrow biopsy and aspiration**

**Samples of bone marrow are removed for testing.**



If blood platelet counts are dangerously low, you may be admitted to the hospital. You'll be treated with an oral or IV steroid. If platelet counts don't improve after 1 to 2 weeks of steroid therapy, you may be treated with rituximab. Or you may be treated with medicines that help raise platelet counts, such as romiplostim (Nplate) or eltrombopag (Alvaiz, Promacta).

## Hyperinflammation

Hyperinflammation is when certain types of white blood cells become overactive and damage the organs and other blood cells in your body. This is a rare but very serious side effect of ICI therapy. Two of these issues can happen at the same time:

- **Cytokine release syndrome**. Cytokines are proteins that carry out different immune-related jobs in the body. Some types cause inflammation. Other types help to reduce it. Too many cytokines in the blood can cause the immune system to become overactive.
- **Hemophagocytic lymphohistiocytosis (HLH)**. HLH can happen when there are low levels of red and white blood cells and platelets. Since it's hard to diagnose, it's sometimes treated as HLH-like syndrome.

Fever is 1 symptom of HLH-like syndrome. You may also have anemia, low blood platelets, and abnormal levels of liver enzymes.

You'll have blood tests to check levels of certain cancer-destroying white blood cells. Your team will test for other things, like infection or cancer of the immune system, that may cause HLH.

**HLH can happen after chimeric antigen receptor (CAR) T-cell therapy.**

For more information on how this side effect is managed, see *NCCN Guidelines for Patients: Immunotherapy Side Effects CAR T-cell therapy*, available at NCCN.org/patientguidelines and on the NCCN Patient Guides for Cancer app.

### Managing HLH-like syndrome

Immunotherapy will be paused. Your care team will likely admit you to the hospital where your treatment will be managed by a hematologist. If possible, your care will be managed by a hematologist who has experience with HLH-like syndrome.

A steroid is the first treatment used for HLH-like syndrome. You'll have 1 of these:

- Prednisone
- Methylprednisolone
- Dexamethasone

Your care team will monitor how certain markers in your blood, including an iron-storage protein, respond to this treatment.

If the steroid doesn't improve hyperinflammation in 5 days, 1 of the following may be added:

- Tocilizumab
- Anakinra (Kineret)
- Ruxolitinib
- Cyclosporine
- Emapalumab

Keep in mind: People with a history of colon disease are at increased risk for a colon tear if taking tocilizumab. Your team will use caution if you take this medicine.

## Key points

- If immunotherapy side effects occur in your blood or bone marrow, a hematologist will likely be involved in your care and treatment.
- Anemia is sometimes treated with a transfusion of donated blood through a vein in the arm.
- You'll likely be treated with steroids if you have a condition that affects your blood or bone marrow.
- Treatment may also include an immunosuppressant or an antibody drug.
- A low level of platelets in the blood is called thrombocytopenia. Treatment may include steroids and intravenous immunoglobulin (IVIG) if there is bleeding.
- You may have a bone marrow biopsy if you have low counts of red blood cells or platelets.
- Hyperinflammatory syndromes are rare but serious side effects. They happen when certain types of white blood cells become overactive and damage other blood cells.

## Questions to ask

- Will I be admitted to the hospital for a blood transfusion?
- What is growth factor support and why do I need it?
- I have had anemia in the past. Does that mean I will have it during ICI treatment?
- Why do I need a bone marrow biopsy and aspiration?
- I have an inherited blood clotting disorder. Will it affect my treatment?

## What's next?

In the next chapter, you can find other resources that may be helpful.

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# Other resources

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90 Questions to ask

Want to learn more? Here's how you can get additional help.

## What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- Specific immune checkpoint inhibitor medicines
- Ways to manage mild symptoms
- Managing fear and anxiety about immune checkpoint inhibitor therapy and possible side effects
- How to talk to the people who support you about the kind of help you may need
- Family planning or fertility

## What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

## Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

**AIM at Melanoma**
aimatmelanoma.org

**AIM at Skin Cancer**
aimatskincancer.org

**Blood Cancer United**
bloodcancerunited.org

**Bone Marrow and Cancer Foundation**
bonemarrow.org

**Cancer*Care***
cancercare.org

**GRACE**
cancergrace.org

**Imerman Angels**
imermanangels.org

**National Leiomyosarcoma Foundation**
nlmsf.org

**NMDP**
nmdp.org

**Save Your Skin Foundation**
saveyourskin.ca

**Triage Cancer**
triagecancer.org

## Questions to ask

- What services are available to me and my caregivers?
- How can I connect with others and build a support system?
- Where can I find help with transportation and childcare?

**Take our survey and help make the NCCN Guidelines for Patients better for everyone!**

NCCN.org/patients/comments

# Words to know

**bananas, rice, applesauce, toast (BRAT)**
A diet recommended for managing symptoms of the lower digestive tract.

**blister**
A fluid-filled sac in the outer layer of skin.

**colitis**
Inflamed bowel (colon).

**computed tomography (CT) scan**

A test that uses many x-rays to make pictures of the inside of the body.

**COX2 inhibitor**
A type of NSAID used to treat inflammation

**CTLA-4**
A protein found on T cells that helps keep the body's immune responses in check.

**dexamethasone**
A steroid given by mouth, usually as a pill.

**diabetic ketoacidosis (DKA)**
When blood sugar becomes dangerously low. Toxic acids (ketones) collect in the blood and urine.

**diarrhea**
Frequent and possibly watery bowel movements.

**disease-modifying anti-rheumatic drug (DMARD)**
A medicine used to treat autoimmune diseases.

**electrocardiogram (ECG)**
A test of the electrical activity of the heart.

**encephalitis**
Inflammation of the brain.

**episcleritis**
Inflammation of the thin outer layer of the eye.

**giant cell arteritis (GCA)**
Inflammation of the lining of the arteries, especially the arteries in the temples.

**Guillain-Barré syndrome**
Rare condition in which the body's immune system attacks the nerves located outside of the brain and spinal cord.

**hematopoietic cell transplant (HCT)**
A treatment that replaces damaged or cells in the bone marrow with healthy cells.

**hemophagocytic lymphohistiocytosis (HLH)**
Low red and white blood cells and platelets.

**hepatitis**
Inflamed liver.

**hypothyroidism**
When the thyroid gland does not make enough thyroid hormone.

**immune checkpoint inhibitor (ICI)**
A type of cancer treatment that blocks contact between immune cells and cancer cells.

**inflammatory arthritis**
A group of disorders caused by immune attack on joints.

**interleukin**
A group of proteins made by white blood cells and other cells in the body. They regulate the body's immune responses.

**intravenous immunoglobulin (IVIG)**
Intravenous infusion of helpful antibodies collected from many donors.

**maculopapular rash**
A rash with both flat patches and bumps.

**meningitis (non-infectious)**
Inflammation of the tissues covering the brain and spinal cord not caused by bacterial infection.

**methylprednisolone**
A steroid given directly into a vein.

**magnetic resonance imaging (MRI) scan**
An imaging test that uses radio waves and strong magnets to view inside of the body.

**myalgia**
Pain in a muscle or group of muscles.

**myasthenia gravis**
Causes weakness in the arms and legs, vision problems, and drooping eyelids or head.

**myocarditis**
A rare but severe side effect in which the heart becomes inflamed.

**myositis**
Weakness, swelling, and/or pain due to inflamed muscles.

**non-steroidal anti-inflammatory drug (NSAID)**
A medicine that decreases fever, swelling, pain, and redness.

**pancreatitis (acute)**
Inflamed pancreas. A rare side effect of immunotherapy.

**pericarditis**
A condition in which the outer lining of the heart becomes inflamed.

**pneumonitis**
Inflammation of 1 or both lungs. The most common symptom is trouble breathing.

**polymyalgia rheumatica (PMR)**
Causes muscle pain and stiffness, especially in the shoulders.

**prednisone**
A steroid given by mouth, usually as a pill.

**proton pump inhibitor (PPI)**
A medicine that reduces stomach acid.

**pruritus**
Itchy skin, with or without a rash.

**steroids**
Drugs that slow the immune system.

**Stevens-Johnson syndrome (SJS)**
A serious skin and mucous membrane condition caused by allergic reactions to certain medicines.

**taper**
Slow stopping of steroids. This is done in an effort to keep side effects from coming back.

**thyroiditis**
Injury to the thyroid gland caused by immune attack. Causes thyroid hormone levels to rise temporarily before dropping too low.

**thyrotoxicosis**
When the body makes too much thyroid hormone.

**toxic epidermal necrolysis (TEN)**
A serious skin and mucous membrane condition caused by allergic reactions to certain medicines.

**transverse myelitis**
A nervous system disorder in which both sides of 1 section of the spinal cord are inflamed.

**tumor necrosis factor (TNF)**
A protein that plays a role in inflammation.

**ultraviolet B (UVB) light therapy**
Use of artificial light on the skin to reduce itching.

**uveitis**
Inflammation of all or part of the middle layer of the wall of the eye (uvea).

**vasculitis**
Inflammation of blood vessels (arteries and veins).

# NCCN Contributors

This patient guide is based on the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Management of Immunotherapy-Related Toxicities, Version 1.2026. It was adapted, reviewed, and published with help from the following people:

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NCCN.org/disclosures.

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Dana-Farber/Brigham and Women's Cancer Center |
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Cancer care recommendations from leading experts at the National Comprehensive Cancer Network® (NCCN®)

# Immunotherapy Side Effects: Immune Checkpoint Inhibitors

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